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BSRNet: a novel neural network for broiler chicken sound signal recognition used in health monitoring

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ABSTRACT

Accurate and effective health monitoring methods for broiler chickens are crucial for the development of modern agriculture and the promotion of agricultural income. Recent studies have primarily focused on broiler sounds, using signal processing algorithms to extract physical information from sound signals, and employing meticulous and specialized extra preprocessing methods to generate high-quality feature datasets. This dataset then serves as the input for a Convolutional Neural Network (CNN) to establish a sound signal recognition model, thereby facilitating indirect health monitoring of broiler chickens. However, the high specialization and complexity of extra preprocessing techniques pose significant challenges to the scalability and universality of practical applications. Moreover, CNNs inherently lack the autonomy to identify core features, preventing the network from effectively modeling using the most distinctive features. Additionally, the model's inability to perform flexible nonlinear mappings based on the specificity of the information captured by the convolutional kernels restricts the network's capability to abstractly represent complex nonlinear relationships and local sound patterns of broiler chickens. These limitations constrain the accuracy of broiler chicken health monitoring. To address these issues, this paper introduces a novel Broiler Sound Recognition Network (BSRNet), which includes innovative components such as Separated Convolution (SepConv) and Adaptive Pre-activated Convolution (APACConv), along with a feature fusion structure (DFF). The proposed SepConv convolution allows the network to autonomously recognize key core features through a feature separation and recombination mechanism, avoiding the limitations associated with cumbersome denoising, filtering, and other extra preprocessing methods. The APACConv, through adaptive pre-activation, performs adaptive nonlinear mappings based on the characteristics of different types of local sound patterns captured by the convolutional kernels, enhancing the network's ability to express complex nonlinear relationships. The DFF structure utilizes both novel convolutions to transform feature channel dimensions (expansion and compression) to achieve multilevel feature fusion and abstract representation. The authors collected a large dataset of broiler chicken sound signals and created four datasets using signal processing algorithms, then conducted comparative experiments with eight methods, achieving competitive results that demonstrate the superiority and advancement of BSRNet. Additionally, ablation experiments validated the effectiveness of BSRNet's core components. Finally, the authors employed T-SNE, Grad-CAM++, and feature visualization techniques for a detailed interpretability analysis of the methods discussed in this paper.

1. Introduction

Health monitoring of broiler chickens holds critical importance in modern agricultural production. With the global population growth and increasing demand for efficient and safe agricultural products, broiler

chickens, as a significant source of meat, play a vital role in ensuring food security and agricultural supply. Diseased broiler chickens directly affect their growth and production performance and may lead to the spread of diseases, potentially causing large-scale epidemics and severe economic losses, thereby posing threats to human health. Therefore,

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accurate and effective health monitoring methods are not only key to enhancing the profitability of broiler farming but also have profound implications for promoting the green development of modern agriculture, increasing agricultural income, and ensuring food safety (Ojo et al., 2022).

Traditional broiler chicken health monitoring primarily relies on biological testing methods, where veterinarians assess the health of the chickens based on biologically manifested symptoms observed manually. However, this diagnostic approach is highly subjective, with accuracy largely dependent on the veterinarian's professional skills and practical experience, making it both time-consuming and challenging to achieve real-time, continuous monitoring (Wu et al., 2022). In contrast to traditional methods, acoustics-based detection technologies offer significant advantages such as non-contact, efficiency, and real-time capabilities. These methods can accurately detect diseases such as Newcastle Disease (ND) (Cuan et al., 2022), bronchitis (Tao et al., 2024) and avian influenza (Cuan et al., 2020). Additionally, they minimize subjective human interference. Consequently, acoustic-based monitoring methods have increasingly become a focal point of research.

Traditionally, acoustic-based monitoring methods utilize signal processing algorithms to extract physical information such as noise intensity and frequency from broiler chicken sound signals in both time and frequency domains. Researchers monitor the health of broiler chickens by analyzing the correlation between physical information and types of sound signals. The prior knowledge embedded in physical information indirectly reflects the physiological state of the broilers, providing a scientifically interpretable basis for researchers to assess their health condition and reducing subjective human interference to some extent. Voslarova et al. (2021) used signal processing algorithms to analyze the noise intensity of sound signals emitted by broilers at different developmental stages, identifying trends in body weight changes and making rough estimations of their health. Yang et al. (2021) applied Fast Fourier Transform to analyze sound signals from broilers of different ages, determining the frequency ranges of six common sound types and establishing the relationship between sound frequency and age, offering a reference for caretakers in monitoring broiler health. However, traditional acoustic monitoring methods still involve researchers or caretakers, not completely eliminating subjectivity due to human factors, leading to errors in broiler health monitoring. Additionally, the relationship between physical information and broiler health often exhibits complex nonlinear characteristics. Manual analysis has limited capability to decipher complex nonlinear relationships effectively, failing to capture and model the intricate dynamics involved, thus impacting the accuracy and reliability of broiler health monitoring.

In recent years, with the rapid development of the field of artificial intelligence, researchers have begun to explore how to combine artificial intelligence algorithms and signal processing techniques to fully utilize the physical information in broiler chicken sound signals for more accurate health monitoring (Tao et al., 2024). Specifically, these methods first use signal processing techniques such as Mel-Frequency Cepstral Coefficients (MFCC) and spectral subtraction to extract acoustical features with physical characteristics, forming feature datasets. Subsequently, machine learning algorithms are employed to learn the complex nonlinear relationships between features and types of sound signals in the feature dataset, entirely avoiding the subjectivity of manual involvement and thereby enhancing the accuracy of broiler health monitoring to some extent. For instance, Rizwan et al. (2016) continuously recorded sound signals from diseased broilers, extracted features using MFCC, and applied Extreme Learning Machine (ELM) and Support Vector Machine (SVM) algorithms to identify clinical signs of broiler health. Sun et al. (2023a) collected sound signals from broilers over different growth periods, then used filtering, pulse extraction, and other methods to construct various feature datasets, training machine learning models for broilers of different ages. Cheng and Zhong. (2015) constructed a feature dataset using MFCC, Linear Predictive Coding (LPC), and Power Normalized Cepstral Coefficients (PNCC), and used

SVM to identify broiler diseases. Liu et al. (2020) constructed a feature dataset from broiler cough sounds through manual selection and trained a model using SVM to accurately identify respiratory diseases in broilers. However, despite the ability of machine learning methods to mitigate the negative impacts of human subjectivity, their capacity for nonlinear fitting is limited (Qin et al., 2020), and the ability to capture the complex nonlinear relationships between acoustic features and sound signal types still requires improvement.

The powerful nonlinear fitting capabilities of Convolutional Neural Networks (CNNs) have become a crucial solution to the issues mentioned above. Researchers have used CNNs to replace machine learning algorithms, fully exploiting the complex nonlinear relationships between physical information and broiler chicken sound signals, and extracting deeper and more abstract feature representations to achieve more accurate health monitoring results. For example, Cuan et al. (2022) extracted spectral features of diseased poultry sound signals using the MFCC method and used these features as inputs to a CNN, proposing a novel Deep Poultry Vocalization Network (DPVN) that significantly enhanced recognition performance. Yuan et al. (2023) developed a method for detecting broiler cough based on vocal features and deep learning, where sound signals were filtered and denoised, and multidimensional features were extracted using MFCC and Fbank (Fbank) before being fed into VGG16 and ResNet18 models for classification, yielding good detection results. Cuan et al. (2020) further proposed a chicken sound recognition method based on CNNs, calculating four major features including time and frequency domains, and combining them to form feature maps as CNN inputs, enhancing the model's robustness and generalizability. Han et al. (2024) utilized the MFCC algorithm to extract a 36-dimensional feature vector from poultry sound signals and used it as input for CNNs to train various poultry gender identification models, achieving high accuracy. Some researchers directly used broiler chicken sound signals as inputs to CNNs (Cuan et al., 2020; Donofre et al., 2020; Sun et al., 2023b). However, these methods lack the introduction of prior knowledge in physical information, affecting model performance and generalizability. Moreover, the original signals contain a large amount of time dimension information, increasing the computational resource burden (Cuan et al., 2020), which is not conducive to practical engineering applications. Therefore, these methods do not perform well in monitoring broiler chicken health.

In summary, the combination of signal processing techniques and artificial intelligence algorithms, with their superior performance, has gradually become the main method for broiler chicken health monitoring. In particular, CNN-based broiler health monitoring methods often achieve higher accuracy than machine learning methods and have shown significant advantages in other areas of poultry health monitoring and behavioral analysis. However, there are still some limitations that restrict the effectiveness of monitoring, as detailed below.

Firstly, although signal processing methods such as MFCC and Short-Time Fourier Transform are effective in extracting sound signal features rich in physical information for CNN inputs, providing necessary prior information for the network, these methods aid in learning, understanding, and establishing the complex nonlinear relationships between broiler acoustic features and categories. However, noise interference exists in broiler sound signals (Sun et al., 2023b), and not all features significantly and critically impact the neural network's modeling process and recognition outcomes. Some irrelevant or redundant features can interfere with the model's learning process, leading to reduced recognition accuracy. As a result, some researchers, in addition to using necessary signal processing algorithms to extract acoustic features, employ extra data preprocessing steps such as filtering, noise reduction, and feature selection (Lv et al., 2023). These steps aim to eliminate the negative impact of noise and irrelevant or redundant features, construct high-quality feature datasets, and build health monitoring models. However, the process of constructing datasets with extra data preprocessing methods is cumbersome and complex, requiring a high level of expertise, which limits its scalability and practical application in real-

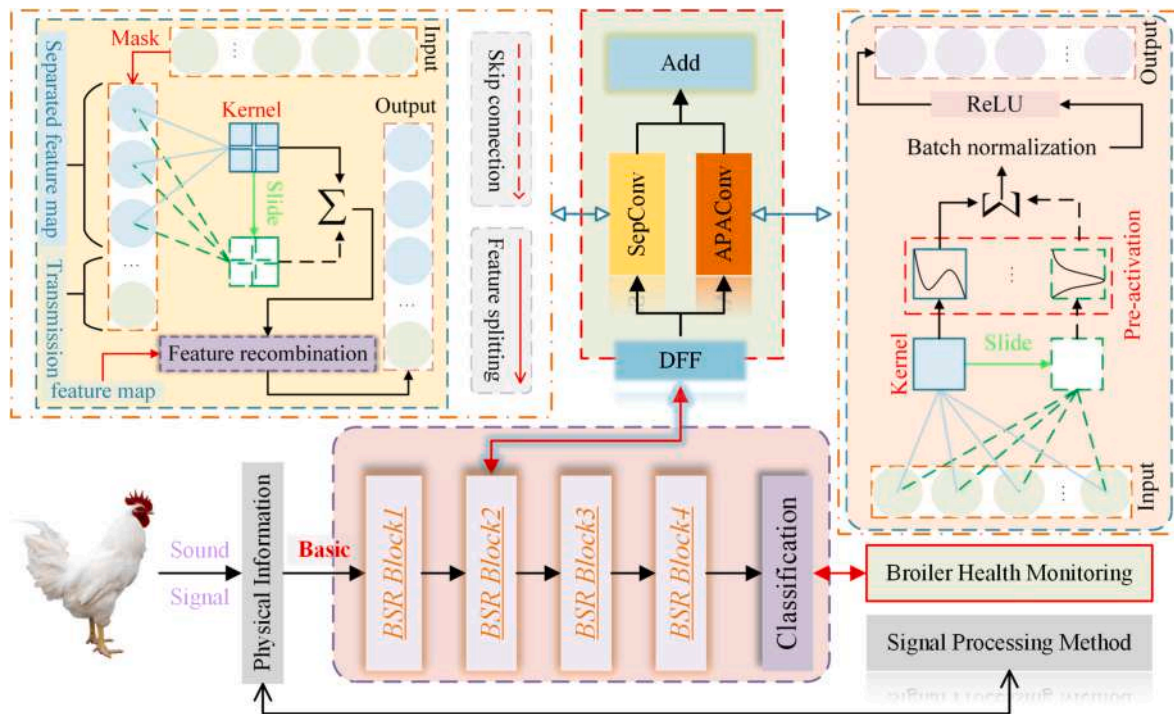


Fig. 1. Schematic Diagram of a New Type of Broiler Sound Signal Recognition Network.

world scenarios.

Secondly, key core features that directly reflect physical information closely related to the health status of broiler chickens are crucial for model building and decision-making. However, CNNs lack the autonomy to identify these core features, preventing the network from effectively utilizing these most discriminative features for modeling, thus posing challenges to the modeling process. Some researchers have used methods such as attention mechanisms to create additional sub-networks within the CNN architecture to autonomously identify core features during the modeling process (Kim et al., 2020). Unfortunately, these methods incur additional computational resources, which also hinder their practical application.

Finally, CNN-based methods for monitoring broiler chicken health often use activation functions (post-activation) after single convolutions or layers composed of multiple convolutions, to enhance the network's capability for nonlinear representation. However, the convolution operation, by sliding the convolutional kernels multiple times, extracts specific local sound patterns that embody distinctiveness. The features contain physical information of varying frequencies and energy amplitudes, which also have potential complex connections. Therefore, using a fixed activation function to uniformly process the complex relationships embedded in various local sound patterns does not allow for flexible nonlinear mapping of the diverse information captured by the convolutional kernels. This limits the network's ability to abstractly represent complex nonlinear relationships and local sound patterns, negatively impacting the accurate identification of broiler chicken sound signal types.

We summarize the limitations of existing studies as follows: (1) Without extra complex data preprocessing, noise in the sound signals, as well as the presence of redundant and irrelevant features in the feature dataset, negatively interfere with the model's learning process. (2) Without the aid of additional sub-networks, the network struggles to autonomously identify key acoustic features that contain physical prior information. (3) Post-activation does not allow for flexible nonlinear mapping based on the characteristics of the information, limiting the network's ability to abstractly represent complex nonlinear relationships and local sound patterns, thereby reducing the effectiveness of the

model in distinguishing broiler chicken sound signal types.

To address the aforementioned issues, this paper proposes a novel broiler chicken recognition network, BSRNet, based on the physical information in broiler chicken sound signals, as illustrated in Fig. 1. BSRNet is a convolutional neural network with four BSR Blocks, capable of autonomously identifying useful core acoustic features and providing greater attention to them. This effectively avoids the interference caused by redundant and irrelevant features and fully utilizes the complex potential nonlinear relationships among the physical information in the diverse local sound patterns extracted by the network's convolutional kernels. This leads to improved accuracy in the diagnosis and recognition of broiler chicken sound signals. This is primarily due to the feature fusion capability of the Dual Path Feature Fusion (DFF) structure used in the BSR Blocks, which includes two innovative types of convolutions: Separated Convolution (SepConv) and Adaptive Pre-activation Convolution (APACConv), that efficiently exploit the physical information in acoustic features. The methods and contributions of this paper are summarized as follows.

- (1) The SepConv convolution proposed in this paper capitalizes on the advantages of gene recombination in sexual reproduction by using a feature separation and recombination mechanism to reduce the dependency among network neurons. This forces the network to autonomously seek core information that positively affects recognition outcomes and assigns higher attention to this information.
- (2) The APACConv convolution, also proposed in this paper, employs adaptive pre-activation to perform adaptive nonlinear mapping based on the characteristics of different types of local sound patterns captured as the convolution kernels slide. This allows the network to fully understand the complex nonlinear relationships between physical information in acoustic features and sound signal types, enhancing the expression ability and flexibility towards different broiler chicken sound signal patterns.
- (3) The DFF structure proposed here consists of a triangular structure that acts at the signal input stage and a bottleneck structure that functions in remaining stages. By transforming feature channel

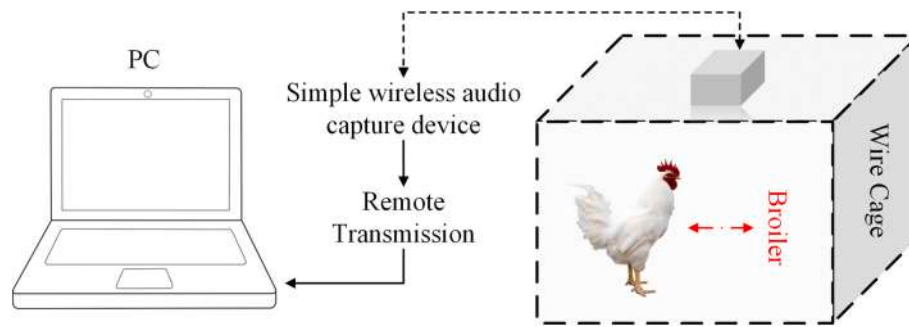


Fig. 2. Schematic Diagram of Equipment Placement.

dimensions (expansion and compression), and combining the core acoustic features extracted by SepConv convolution with the diverse complex nonlinear local features captured by APAConv convolution, it achieves multilevel feature fusion and abstract representation.

- (4) BSRNet, utilizing the aforementioned innovative components, can autonomously capture useful features and enhance the diversity of feature extraction, ultimately improving the accuracy of broiler chicken sound signal recognition. Notably, the autonomy in capturing useful features in this method derives from the network itself, rather than using traditional attention mechanisms or complex data cleansing, which enhances the practicality of this method. Furthermore, the innovative components included in this method can not only cooperate with each other but also be flexibly integrated independently with other acoustic recognition tasks.
- (5) To validate the effectiveness and advancement of the method, we collected a large number of broiler chicken sound signals and constructed several acoustic feature datasets with physical information through signal processing methods, followed by comprehensive comparative and ablation experiments. The competitive experimental results obtained demonstrate that our method can effectively enhance the accuracy of broiler chicken sound signal recognition, achieving accurate broiler chicken health monitoring.

The rest of this paper is organized as follows. Section 2 first introduces the collection and gathering process of broiler chicken sound signals, as well as the production process of the acoustic feature datasets, then details the BSRNet neural network structure and the design processes of the DFF structure, SepConv convolution, and APAConv convolution. In Section 3, the authors conduct extensive and detailed comparative and ablation experiments with eight methods to validate and analyze the effectiveness of BSRNet. Additionally, we employ T-SNE (Van Der Maaten, 2014), Grad-CAM++ technologies (Zhang et al., 2022), and feature visualization methods for an in-depth interpretability analysis of BSRNet. Finally, in Section 4, we summarize the work of this paper, analyze its limitations, and provide future perspectives.

2. Materials and methods

2.1. Dataset sources

2.1.1. Data collection

The data for this study were collected from the Huixing Farm in Linkou County, Mudanjiang City, Heilongjiang Province, China (44°91'N, 130°02'E). The farm houses approximately 8400 Arbor Acres broilers in Greenhouse No. 1 (Sun et al., 2023a), distributed across 35 broiler farming areas arranged in an east–west direction. Each broiler farming area was divided into multiple square sub-areas, each with a length and width of 2 m, separated by wire fences. Approximately 30

broilers were raised in each sub-area. It is important to note that the broiler growth cycle lasts about 40 days. During this period, there are differences in health status and behavioral performance at different growth stages. Furthermore, through communication with veterinary husbandry experts, the authors learned that broilers at the submature stage around day 30 and at the mature stage around day 40 have critical health conditions that are essential for management. To enhance data diversity and ensure coverage of the acoustic feature variations of broilers from the submature to mature stages, and to provide the necessary data foundation for model development, this study used broilers from day 29, 30, and 31 (submature stage) and day 38, 39, and 40 (mature stage) as the subjects of the research. This experiment has been approved by the Laboratory Animal Ethics Committee of Heilongjiang University. The approval number is 20,220,811,007 and the approval date is August 18, 2022.

In this study, sound signals were collected using a simple wireless audio acquisition device (hereafter referred to as the device), which includes a microphone, a wireless transmission module, and a remote PC (Sun et al., 2023b). During operation, the device records sound using a high-precision microphone with a sampling rate range of 200 Hz to 10,000 Hz and a sensitivity of 12 mV/Pa. The recorded sounds include those reflecting the health status of both healthy and diseased broilers at the submature and mature stages, as well as other sounds such as those produced by the movement of caretakers. These recorded sounds are then transmitted to the remote PC via the wireless transmission module and saved in WAV format.

The specific collection process was as follows: Within Greenhouse No. 1, three sub-areas not used for broiler chicken breeding (hereafter referred to as data collection sub-areas) were selected. In each data collection sub-area, a wire mesh cage (1 m in length, width, and height) was placed to facilitate the placement of the audio acquisition device, while also preventing overlapping sound signals from multiple sick broilers, which could affect data accuracy. A diagram of the device placement is shown in Fig. 2. Subsequently, 30 sick broilers were selected from within Greenhouse No. 1, with 10 placed in each data collection sub-area. The chickens were kept in the three data collection sub-areas for a period to acclimatize to the environment. During data collection, the device was positioned directly above the center of the cage's wire mesh. A caretaker then took the first sick broiler from data collection sub-area 1, placed it in the cage, and recorded sound data for 10 min. After recording, the chicken was taken to a medical room for treatment. After waiting 10 min, the second sick broiler was placed in the cage, and the process was repeated until all 10 sick broilers had been individually placed in the cage of data collection sub-area 1, completing one round of data collection experiments. The same procedure was then repeated in data collection sub-areas 2 and 3. It is important to emphasize that the diseased broilers were diagnosed with bronchitis by veterinary husbandry experts at the early stage of illness, and after the signal collection, all diseased broilers received timely treatment and their health was not negatively affected. After collecting the sound signals of the diseased broilers, 30 healthy broilers were selected in the

Table 1

Broiler Sound Signal Types. “SMS-Cough” and “MS-Cough” represent the cough sounds in the submature and mature stages, respectively. “SMS-Flapping” and “MS-Flapping” represent the wing-flapping sounds in the submature and mature stages, respectively.

Sound Signal Type	Label	Growth Stage
MS-Cough	0	Mature Stage
MS-Flapping	1	
Rale	2	
Snore	3	Submature Stage
Chirping	4	
SMS-Cough	5	
SMS-Flapping	6	
Grunting	7	

same way from the greenhouse, with 10 broilers placed in each data collection sub-area, and the audio signal recording was conducted following the same steps as outlined above. A total of seven experiments were conducted, with each experiment using broilers that had not previously been recorded. In total, 420 broilers’ audio signals were collected, including 210 healthy broilers and 210 diseased broilers.

2.1.2. Dataset construction

In Section 2.1.1, the authors recorded the audio signals of healthy and diseased broilers. However, due to the inevitable inclusion of background sounds from the work of the caretakers and other noise during the recording process, the author segmented the recorded audio signals to isolate those representing different types of vocalizations from the broilers. During this process, multiple discussions were held with veterinary husbandry experts, who listened to each audio segment. Based on their professional judgment and proofread, we selected and retained the most certain audio signal types that best reflect the health status of broilers at different stages of growth, at the same time, the corresponding types of the vocal signals were labeled, providing more representative data for the study. For those audio signals that the experts could not definitively identify, considering that the uncertainty of these signals might interfere with the stability and accuracy of model training, the authors decided not to include them in the model training process. Given that, within each stage, the vocal differences of the broilers in adjacent growth days are relatively small, and there are certain physiological and behavioral differences in the vocal characteristics between the two stages. Therefore, the authors combined segments of sound signals of the same type from both stages, which not only resulted in a

more abundant amount of data, but also helped to enhance the model’s generalization ability. Additionally, the authors treated the audio signals from the two stages as independent health status types to ensure the diversity of health conditions. Ultimately, for the submature stage broilers, the authors retained the sounds of diseased broilers Chirping, Cough, and Grunting, as well as the sounds of healthy broilers flapping their wings. For the mature stage broilers, the retained sounds included Cough, Rale, and Snore from diseased broilers, as well as the sound of healthy broilers flapping their wings. In total, eight types of audio signals were identified, with corresponding labels as shown in Table 1.

In fact, broilers at the mature stage have more developed vocal cords compared to those at the submature stage, resulting in sound signals with greater amplitude. Additionally, during the submature stage, broiler chirping tends to be rapid and high-pitched, cough is relatively prolonged, grunting is typically low and irregular, and wing-flapping sounds are of longer duration and higher amplitude. In contrast, broilers at the mature stage exhibit rale that is sharper and more prolonged, often accompanied by noticeable airway obstruction or mucus accumulation. Their cough becomes more frequent and intense, and the overall amplitude of snore increases significantly, characterized by deeper and more sustained sounds. Moreover, wing-flapping sounds become more forceful. Overall, the vocal signals emitted by healthy broilers exhibit stable amplitude characteristics with certain periodicity, reflecting the regular and normal physiological state of healthy signs. On the other hand, sick broilers often display irregular, sudden, chaotic, or severe abnormal fluctuations, reflecting pathological states associated with respiratory system disorders. The waveform plots of the eight signal types shown in Fig. 3 exhibit signal characteristics consistent with the above descriptions.

Considering the variations in amplitude among these sound signal segments, the authors normalized these eight types of sound signals (Wang et al., 2024), as shown in Eq. (1), where x represents the sound signal segment, μ represents the mean, σ represents the standard deviation, and y represents the normalized sound signal.

$$y = \frac{x - \mu}{\sigma} \quad (1)$$

Subsequently, in order to extract the physical information from the original sound signals of the eight categories of broilers and introduce the necessary prior information for the subsequent CNN. Furthermore, acoustic feature datasets with different characteristics can test the generalization capability of BSRNet, while different signal processing algorithms often yield datasets with distinct features. Therefore, the

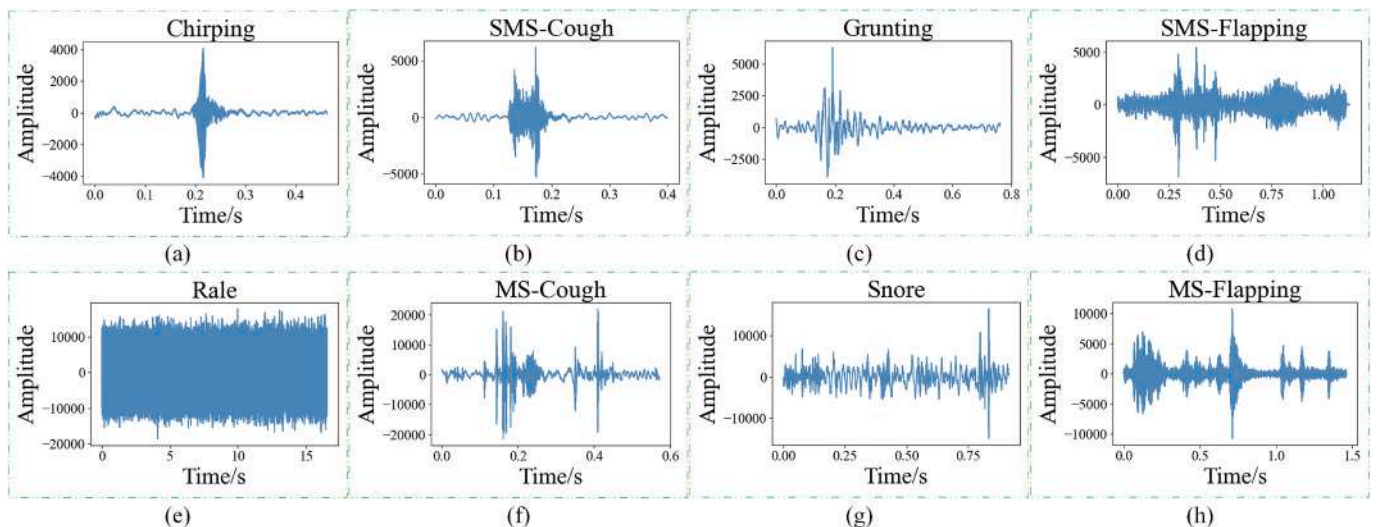


Fig. 3. Waveforms of the 8 types of sound signals. “SMS-Cough” and “MS-Cough” represent the cough sounds in the submature and mature stages, respectively. “SMS-Flapping” and “MS-Flapping” represent the wing-flapping sounds in the submature and mature stages, respectively.

Table 2

Characteristics of the acoustic feature datasets. The second column indicates the signal processing algorithm used to construct the acoustic feature dataset.

Dataset	Algorithm	Dimension	Dataset Characteristics
RJ1	FBank	80	Reflects the original spectral information of the sound signal.
RJ2	Spectrogram	201	Provides finer frequency-domain details for each time segment, better representing temporal variations in the audio signal; typically has high acoustic feature dimensionality.
RJ3	Mel Spectrogram	128	Emphasizes low-frequency information and contains richer time–frequency features.
RJ4	MFCC	40	Compresses features by applying discrete cosine transform to the Mel spectrum, offering low-dimensional and discriminative features.

Table 3

Detailed description of the sample size in the feature datasets.

Dataset	Training Set	Test Set	Total
RJ1	13,048	5593	18,641
RJ2	12,140	5203	17,343
RJ3	12,280	5263	17,543
RJ4	11,487	4924	16,411

authors employed four conventional signal processing algorithms—Fbank (Wu et al., 2023), Spectrogram (Léonard, 2007), Mel Spectrogram (Shen et al., 2018), and MFCC (Ai et al., 2012)—to process the audio signals of eight categories, resulting in four distinct acoustic feature datasets. These datasets were then split into training and testing sets in a 7:3 ratio. Subsequent comparative experiments, ablation studies, and interpretability analysis will be conducted on these four datasets. The characteristics and sample sizes of the datasets are detailed in Table 2 and Table 3, respectively.

In fact, the cumbersome processing of original audio data increases the difficulty of practical application and is not conducive to replication

by subsequent researchers. Therefore, the dataset construction process in this study is relatively simple, involving only necessary signal segmentation, normalization, and acoustic feature extraction, before serving as inputs for the subsequent BSRNet to validate the method's effectiveness. It should be noted that, to maximize the effectiveness of acoustic feature extraction and ensure the fairness of the research results, the hyperparameter settings for the four signal processing algorithms used in this study strictly follow the original algorithm configurations.

2.2. BSRNet overall structure

This paper proposes a novel broiler chicken sound signal recognition network (BSRNet), which autonomously identifies useful core features and dynamically enhances focus on these core features while simultaneously diminishing attention to redundant, irrelevant, and noise features. Additionally, it performs targeted and flexible nonlinear mapping by adaptively pre-activating based on the characteristics of various local sound patterns extracted by the convolutional kernels, thereby enhancing the diversity and richness of feature extraction. Therefore, the proposed method only requires signal processing algorithms to introduce necessary physical prior information, without the need for extra cumbersome preprocessing (such as filtering, denoising, and feature selection) of the raw sound signals. This effectively improves the accuracy of broiler health monitoring and the widespread applicability of the method. The overall structure is shown in Fig. 4, and detailed parameters are summarized in Table 5 of Section 3.1.

BSRNet initially processes the input through four Blocks for layer-by-layer feature extraction. Each BSR Block consists of a Stage layer and a Global Max Pooling (GMP) layer (LeCun et al., 1998), where GMP is used for downsampling the features to further highlight key features and enhance the model's sensitivity to important characteristics. The Stage layer is composed of the proposed SepConv convolution and APAConv convolution, which effectively capture core features and provide a more abstract and precise nonlinear representation. It also learns the local details and global trends of various local sound patterns, followed by the fusion of feature information extracted by the two novel convolution operations through the DFF structure. After the four BSR Blocks, a

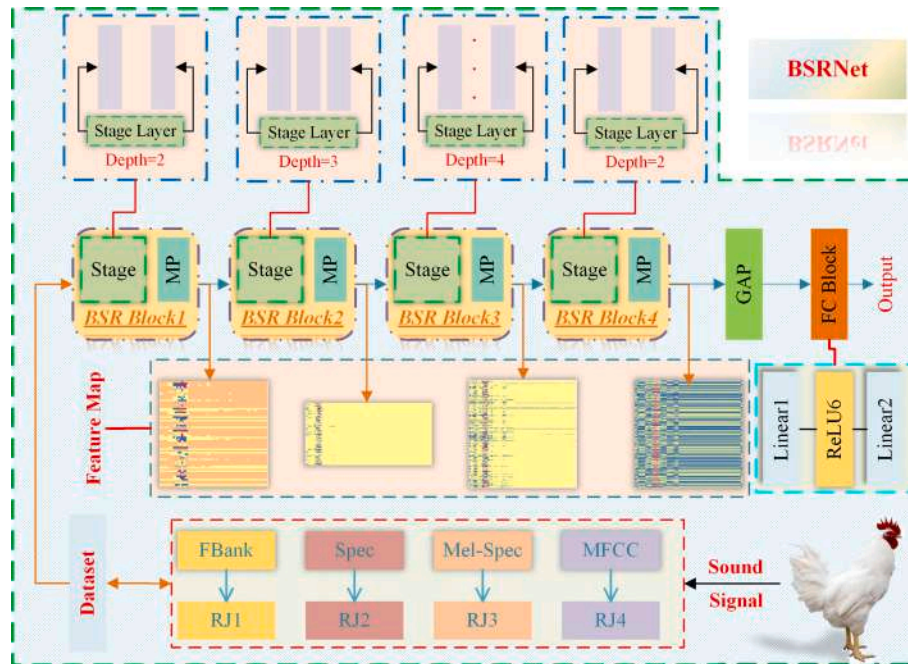


Fig. 4. Schematic Diagram of the Overall Structure of BSRNet. “Spec” and “Mel-Spec” represent the datasets RJ2 and RJ3, which are constructed using the Spectrogram algorithm and the Mel Spectrogram algorithm, respectively.

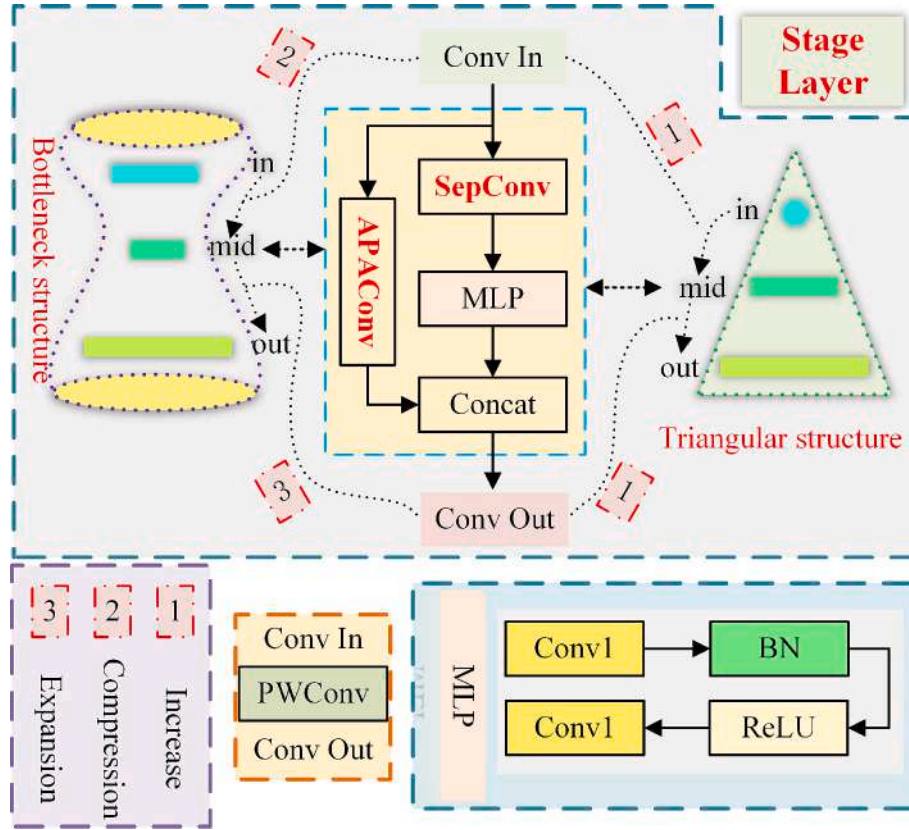


Fig. 5. DFF Structure Diagram.

Global Average Pooling layer (Szegedy et al., 2015) is used for spatial dimensionality reduction of the feature maps, aggregating the spatial information of each feature map into a single numerical representation, enhancing the global nature of feature expression while preventing overfitting. The features processed by the global average pooling are then fed into the classification block (FC Block). The FC Block consists of two fully connected layers (Krizhevsky et al., 2012) and the activation function ReLU6 (Sandler et al., 2018), designed to integrate and map the extracted features, linking them to the sound signal categories, ultimately completing the recognition of broiler chicken sound signals and achieving broiler chicken health monitoring.

2.3. Stage layer

The Dual Path Feature Fusion structure (DFF structure) proposed in this paper is used to form the Stage layer. It transforms the channel dimensions of the input feature maps and combines the two novel convolutions proposed in this paper to achieve multi-level feature fusion and expression. The schematic diagram of the calculation is shown in Fig. 5. The DFF structure primarily consists of a triangular structure at the BSRNet input stage (Block 1) and a bottleneck structure at the subsequent stages of the network (Block 2 to Block 4), with each structure comprising three stages: the first, second, and third stages, where the channel dimensions at each stage are defined as *in*, *mid*, and *out*, respectively. The detailed design process of these two configurations is as follows.

The triangular structure consists of three stages, with the relationship between the channel dimensions at each stage shown in Eq. (2), and the overall computation process detailed in Eq. (3), where $PWConv(\bullet)$ denotes the pointwise convolution operation, x_{stage} denotes the input of the Stage layer, y_{stage} denotes the output of the Stage layer. Since the initial channel dimension of the input data is 1 (i.e., $in = 1$), in order to utilize the neural network's capability to extract features across multiple

channels in different dimensions, it is necessary to expand the channel dimensions of the feature map. Here, the authors use pointwise convolution (Chollet, 2017) in the first stage to expand the channel dimensions from the original *in* to *mid*.

$$in < mid < out \quad (2)$$

$$\begin{cases} x' = PWConv(x_{stage}) \\ x_1 = x' \\ x_2 = x' \\ y_1 = MLP(SepConv(x_1)) \\ y_2 = APACConv(x_2) \\ y_3 = Concat(y_1, y_2) \\ y_{stage} = PWConv(y_3) \end{cases} \quad (3)$$

For the second stage, the authors perform feature extraction on the feature maps after dimension expansion through SepConv convolution and APACConv convolution, respectively. To thoroughly capture the nonlinear relationships between the physical information in acoustic features and sound types, as well as the potential correlations among physical information, the authors use a parallel dual-branch structure to fuse the multi-perspective features extracted by the two types of convolutions. This enhances the model's robustness and generalization ability in recognizing different sound types. Finally, in the third stage, pointwise convolution (PWConv) is used again to expand the feature dimensions from *mid* to *out*, further enhancing the model's expressive capability and the diversity of features.

It is noteworthy that in the branch containing the SepConv convolution, the authors use an MLP structure following the SepConv convolution to capture complex interactions globally within the feature space, further enhancing feature expression capability. Unlike traditional MLP structures, we replace fully connected layers with standard convolution operations using a kernel size of 1 to perform multilayer nonlinear combinations, enhancing the abstract representation of implicit corre-

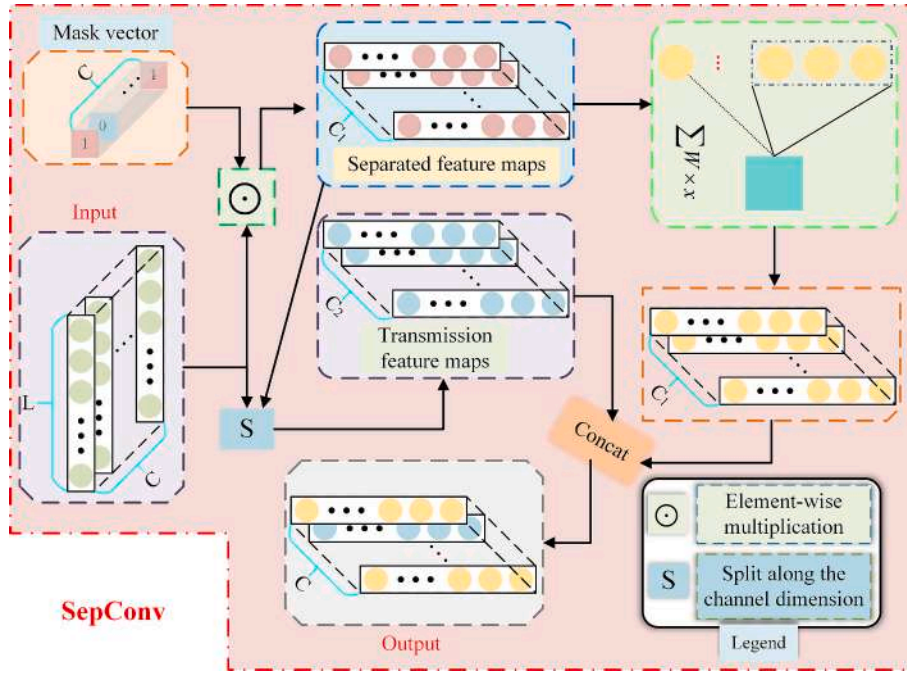


Fig. 6. Diagram of Separated Convolution.

lations within the features. The MLP structure is designed to facilitate global feature interactions and reorganization. The combination of these elements further improves the model's precision in autonomously capturing useful core features. The MLP structure is described in Eq. (4), where $\text{Conv}_1(\bullet)$ represents standard convolution operations with a kernel size of 1, $\sigma(\bullet)$ represents the activation function, and $\text{BN}(\bullet)$ denotes batch normalization, x_m denotes the input of the MLP structure.

$$\text{MLP}(x_m) = \text{Conv}_1(\sigma(\text{BN}(\text{Conv}_1(x_m)))) \quad (4)$$

The computational process of the bottleneck structure is similar to that of the triangular structure, with the difference lying in how the triangular structure gradually expands the feature map channels from the input stage to fully utilize the rich expressive capability of multiple channels, enhancing the capture and representation of complex features. In contrast, the bottleneck structure achieves feature compression and reconstruction by compressing and expanding the feature channel dimensions, thoroughly exploring the potential differences and associations within the acoustic features in each Stage layer, effectively enhancing the model's global feature fusion and expressive capabilities. This provides meaningful information support for accurate decision-making by the model, improving the compactness and richness of feature expression. The relationships between the input, middle, and output feature map channel dimensions in the bottleneck structure are shown in Eq. (5).

$$\text{mid} < \text{in} < \text{out} \quad (5)$$

Next, the authors will detail the design processes for SepConv convolution and APACConv convolution.

2.4. Separated convolution

To endow the network with autonomy in recognizing useful core features and to avoid interference from redundant or irrelevant features during the learning process, this paper proposes a novel SepConv convolution inspired by the advantages and concepts of sexual reproduction genetic recombination. The core steps include feature splitting and feature recombination, with a computational diagram shown in Fig. 6.

Given the input feature map $x_{\text{sep}} \in \mathbb{R}^{C \times L}$ for the SepConv convolu-

tion, where C represents the number of channels and L represents the feature dimensions. The feature splitting step unfolds along the channel dimension of the feature map, dividing the input into two parts for the SepConv convolution. For clarity, during the feature splitting process, the authors refer to the feature map formed by channels autonomously selected by neurons with a certain probability as the '**separated feature maps**', while the remaining channels form the '**transmission feature maps**'.

Initially, a mask vector m with the same channel dimension as the input x_{sep} is generated to form the separated feature map. Here, $m \in \mathbb{R}^{C \times 1}$ and $m = [m_1, m_2, m_3, \dots, m_i, \dots, m_C]$, is a binary vector composed of elements following the same Bernoulli probability distribution (1 indicates selection, 0 indicates no selection). Element m_i represents whether the i^{th} channel of x_{sep} is selected, drawn as a binary variable from a Bernoulli distribution, as detailed in Eq. (6). Here, $p = 1/\tilde{c}$ represents the probability of selecting channel i of the input feature map, with $\tilde{c}$ being a hyperparameter that controls p , referred to as the separation number (SN).

$$m_i \sim \text{Bernoulli}(p) \quad (6)$$

Then, the mask vector m and the input feature map x_{sep} are multiplied channel-wise to form the masked feature map $\tilde{x}_{\text{sep}}$, as shown in Eq. (7).

$$\tilde{x}_{\text{sep}} = x_{\text{sep}} \odot m \quad (7)$$

Herein, $\odot$ represents channel-wise multiplication, referring specifically to applying the mask to each channel of the input feature map x_{sep} , i.e., retaining the channels where the mask is 1 and setting the values of channels where the mask is 0 to zero. Subsequently, the feature map composed of non-zero channels is extracted to form the separated feature map x_{sf} , as shown in Eq. (8), where $\text{Select}(\bullet)$ denotes the extraction operation. Then, the separated feature map x_{sf} is removed from the input feature map x_{sep} along the channel dimension, resulting in the transmission feature map x_{tf} , as illustrated in Eq. (9).

$$x_{\text{sf}} = \text{Select}(\tilde{x}_{\text{sep}}) \quad (8)$$

$$x_{\text{tf}} = x_{\text{sep}} - x_{\text{sf}} \quad (9)$$

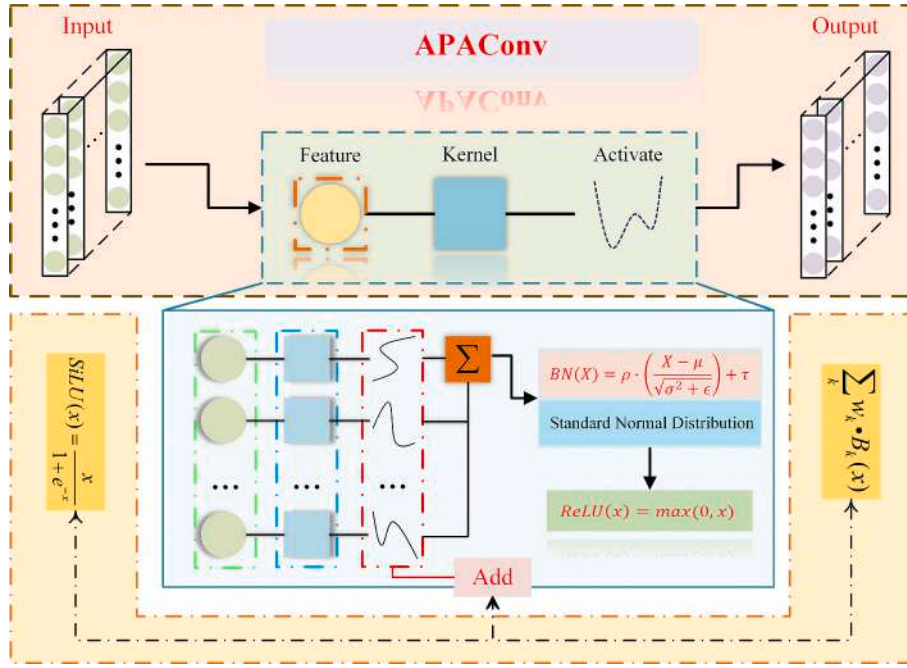


Fig. 7. Diagram of APAConv Convolution.

After obtaining the separated feature map and transmission feature map, a feature recombination step is conducted. Specifically, the authors extract features from the separated feature map using a small kernel filter (kernel size of 3) to form a differentiated feature map y_{sf} . Simultaneously, the transmitted feature map is directly transferred through a skip connection, forming feature map y_{tf} . The separated and transmission feature maps are then concatenated along the channel dimension to form the output y_{sep} of the SepConv convolution. The specific process is shown in Eq. (10), where $Kernel(\bullet)$ denotes the small kernel filter, and W represents the convolutional kernel weights.

$$\begin{cases} y_{sf} = Kernel(W, x_{sf}) \\ y_{tf} = x_{tf} \\ y_{sep} = y_{sf} + y_{tf} \end{cases} \quad (10)$$

In conclusion, when SepConv convolution is used as the feature extraction layer of the network, there is no need for extra complex preprocessing (such as filtering, denoising, or feature selection) of the raw audio signals. Specifically, as the network deepens, each use of SepConv convolution randomly and spontaneously selects and recombines the feature maps along the channel dimension. This prevents the convolutional layers from depending on a fixed set of neurons for feature extraction, requiring each neuron to independently and spontaneously learn features, thus breaking the strong dependencies between neurons and enhancing the overall accuracy and robustness of the neural network. Moreover, this autonomy in capturing useful features is inherent and does not rely on subnetworks to generate additional weights, nor does it require additional computational resources, making it more suitable for practical industrial applications.

2.5. Adaptive pre-activation convolution

To address the limitations of post-activation, which cannot flexibly perform nonlinear mappings based on the diverse local abstract sound patterns' properties extracted by multiple convolution kernels, thus restricting the network's ability to abstractly represent complex nonlinear relationships and local sound patterns and reducing the richness and precision of the network's feature extraction, this paper proposes an Adaptive Pre-activation Convolution (APAConv Convolution). The structural schematic of APAConv Convolution is shown in

Fig. 7.

The APAConv convolution first uses a kernel of size 1 to learn the rich feature information in the input feature map without altering the information in the feature space. It utilizes an adaptive pre-activation function to flexibly perform nonlinear mapping on the diverse information captured by each convolution kernel, extracting more refined and rich feature information. Subsequently, it fuses the multi-scale, multi-perspective feature information extracted by multiple convolution kernels to enhance the diversity and completeness of feature expression. This approach is markedly different from the traditional post-activation method, which uses convolution followed by a fixed activation function. Finally, batch normalization (BN layer) (Ioffe, 2015) is applied to normalize the mean and variance of the features, reducing shifts in feature distribution, and a ReLU activation function (Nair and Hinton, 2010) is used for global nonlinear transformations, further enhancing the extraction and representation of complex features and suppressing the impact of negative values to avoid the problem of gradient vanishing.

$$\begin{cases} APAConv(x_{apa}) = \sigma \left(BN \left(\sum_{d=1}^C \sum_{l=1}^L pr_{d,l} \left(Kernel_{d,l}^1(x_{apa(d,l)}) \right) \right) \right) \\ BN(x) = \rho \cdot \left(\frac{x - \mu}{\sqrt{\lambda^2 + \epsilon}} \right) + \tau \end{cases} \quad (11)$$

The specific definition is as follows: the authors define the input feature map for the APAConv convolution as $x_{apa} \in \mathbb{R}^{C \times L}$, where C represents the number of channels in the feature map and L represents the feature dimensions. The definition of the APAConv convolution is shown in Eq. (11). Here, $pr(\bullet)$ represents the adaptive pre-activation function for each convolution kernel, $Kernel^1(\bullet)$ denotes a convolution kernel of size 1, and $\sigma(\bullet)$ represents the ReLU activation function. For the BN layer, ρ and τ respectively denote the learnable scaling and shifting parameters, x represents the input to the BN operation, μ represents the mean of x , λ^2 represents the variance of x , and ϵ represents the smoothing factor.

The adaptive pre-activation function $pr(\bullet)$ is designed to ensure flexible nonlinear mapping of distinct feature information extracted by each convolution kernel. It is composed of the Basic Spline function (BS

function) and the SiLU function, as defined in Eq. (12). Research (Fakhoury et al., 2022) indicates that the curve shape and smoothness of the BS function are highly flexible and adaptable, making it widely used in scenarios that require complex smooth curves, such as drone path planning. For this purpose, the authors utilize a neural network to train the learnable parameter w_k , and, combined with the local fitting properties of $B_k(x_{pr})$, control the smoothness of the curve and its local impact range to perform targeted nonlinear mapping based on the characteristics of different information, as shown in Eq. (13), x_{pr} denotes the input of the pre-activation function. Additionally, to further enhance the feature expression capability, the authors also use the SiLU activation function (Elfwing et al., 2018), leveraging its self-gating properties to adjust the activation intensity, enhancing the adaptability of the adaptive activation function, as indicated in Eq. (14). Finally, after fusing the activation results of both functions, the BN layer and ReLU function are used to perform a global nonlinear mapping of features, while reducing shifts in feature distribution.

$$pr(x_{pr}) = BS(x_{pr}) + SiLU(x_{pr}) \quad (12)$$

$$BS(x_{pr}) = \sum_k w_k \cdot B_k(x_{pr}) \quad (13)$$

$$SiLU(x_{pr}) = \frac{x_{pr}}{1 + e^{-x_{pr}}} \quad (14)$$

2.6. Interpretability analysis

The authors conduct a detailed interpretability analysis of BSRNet. Specifically, the Grad-CAM++ technique is utilized to visualize the neural network's focus on input data and its association with different types of sound signals, confirming the network's autonomous perception and capture of core features, as well as its ability to dynamically differentiate between irrelevant or redundant information. Additionally, the T-SNE technique is used to analyze the feature extraction and discrimination performance within each Block of BSRNet, further illustrating the hierarchical representation capabilities of the network structure. Lastly, the output feature maps from each Block are visualized to intuitively display the feature extraction effects of different Blocks.

2.7. Evaluation metrics

To validate the effectiveness of the methods and core components proposed in this paper, the authors employ four evaluation metrics: accuracy (Acc), micro precision (MP), micro recall (MR), and micro F1 score (MF1). These metrics are used in comprehensive comparative and ablation experiments against several recent methods. The definitions of these evaluation metrics are shown in Eq. (15), Eq. (16), Eq. (17), and Eq. (18), respectively.

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

$$MP = \frac{TP}{TP + FP} \quad (16)$$

$$MR = \frac{TP}{TP + FN} \quad (17)$$

$$MF1 = \frac{2TP}{2TP + FP + FN} \quad (18)$$

3. Results and discussion

This section experimentally validates the effectiveness and advanced nature of the novel broiler chicken sound signal recognition neural network, BSRNet, proposed in this paper. The authors first describe the experimental setup and the selection of optimal hyperparameters. Then,

Table 4

Experimental hyperparameter settings.

Framework	Programming language type and version	Optimizer	Learning rate	Batch Size
Pytorch	Python3.9	Adam	0.0001	32

Table 5

Detailed parameter settings for the BSRNet network structure. *Depth* denotes the number of stacking iterations of the Stage layer. *cin*, *mid*, and *cout* denote the number of channel dimensions in the three stages of the DFF structure, respectively, and *SN* denotes the number of separations in the SepConv convolution. *C* denotes the number of sound types. *L* denotes the dimension of the input data. *B* represents Batch Size.

Blocks	Stage layers	Parameters	Feature shape
Input	—	—	$B \times 1 \times L$
BSR Block 1	Stage1	Depth = 2, cin = 1, mid = 16, cout = 64, SN = 3	$B \times 64 \times L$
	MaxPool_1D	L	
BSR Block 2	Stage2	Depth = 3, cin = 64, mid = 32, cout = 128, SN = 3	$B \times 128 \times L$
	MaxPool_1D	L	
BSR Block 3	Stage3	Depth = 4, cin = 128, mid = 64, cout = 256, SN = 3	$B \times 256 \times L$
	MaxPool_1D	L	
BSR Block 4	Stage4	Depth = 2, cin = 256, mid = 128, cout = 512, SN = 3	$B \times 512 \times L$
	MaxPool_1D	L	
AdaptiveAvgPool_1D	—	1	$B \times 512 \times 1$
Flatten	—	—	$B \times 512$
FC Block	Linear1	(512, 256)	$B \times 256$
	ReLU6	—	$B \times 256$
	Linear2	(256, C)	$B \times C$

they conduct a comprehensive evaluation of the method through ablation studies and comparative experiments. Following this, the interpretability of BSRNet is analyzed using methods such as T-SNE, Grad-CAM++, and feature visualization techniques.

3.1. Experimental parameter settings

The experiments were conducted on hardware equipped with an NVIDIA RTX 4090D GPU. The model was trained using the hyperparameters listed in Table 4, and the results on the test set are summarized. To enhance the reproducibility of the results, the model parameters for BSRNet are detailed in Table 5.

3.2. Ablation studies

This subsection verifies the effectiveness of the two novel

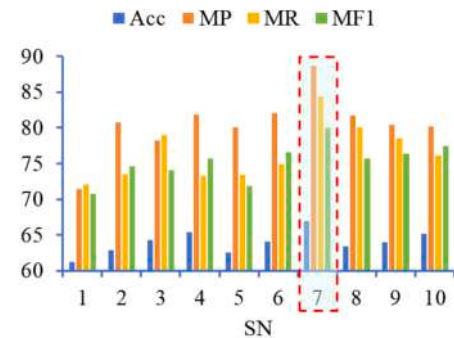


Fig. 8. Comparison of experimental results with different separation numbers on the RJ1 dataset.

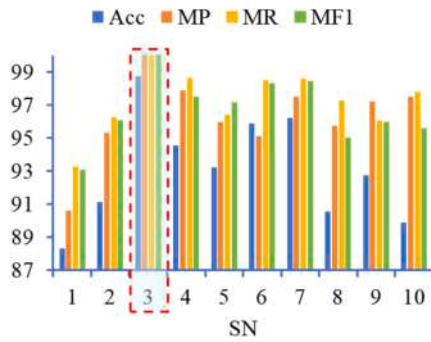


Fig. 9. Comparison of experimental results with different separation numbers on the RJ2 dataset.

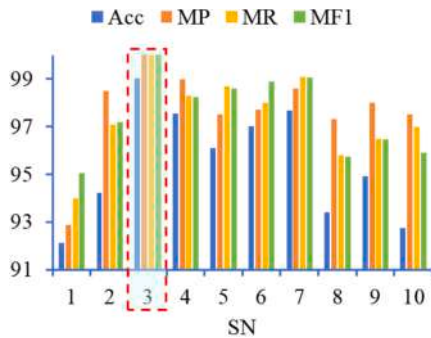


Fig. 10. Comparison of experimental results with different separation numbers on the RJ3 dataset.

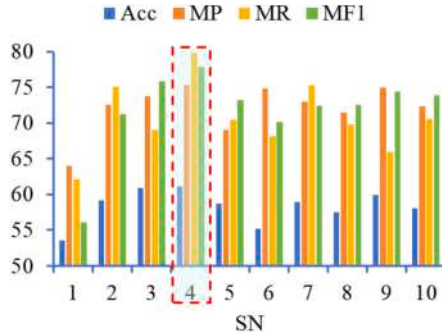


Fig. 11. Comparison of experimental results with different separation numbers on the RJ4 dataset.

convolutions proposed in this paper through ablation studies. The separation number is a hyperparameter of the SepConv convolution, which determines the channel dimensions of the separated feature map and has a significant impact on the network's autonomy in capturing core features. To ensure that BSRNet achieves optimal accuracy in recognizing broiler chicken sound signals, facilitating subsequent experimental analysis, the authors conduct performance tests on the four datasets by setting different separation numbers. The experimental results are shown in Fig. 8, Fig. 9, Fig. 10 and Fig. 11.

In fact, when the separation number is set to 1, the channel dimensions of the separated feature map are the same as those of the input feature map, resulting in the channel dimensions of the transmission feature map being zero, essentially eliminating the transmission feature map and directly extracting features from the entire input via the convolution kernel. Therefore, at this point, the SepConv convolution is equivalent to a standard convolution. It can be observed that the performance of the neural network does not reach the optimal level across

the four datasets, and the classification performance with a separation number of 1 is lower in most cases compared to other separation number settings. This indicates that compared to traditional standard convolution, SepConv convolution has certain advantages, enabling more effective enhancement of the neural network's feature extraction capabilities and overall performance.

If the transmission feature map is not considered and only the separated feature map is analyzed, increasing the separation number would theoretically further reduce the richness of the features. Thus, as the separation number increases, the performance of the neural network should theoretically show a regular declining trend. However, the experimental results indicate that although the performance of the neural network varies with different separation numbers, its accuracy does not decrease gradually but fluctuates up and down. In most cases, it still surpasses the performance of networks using standard convolution. This suggests that the feature recombination mechanism, designed using the concept of skip connections, ensures the integrity of the channel information in the feature maps during transmission. This integrity provides the network with greater robustness, preventing the separation number changes from having too significant a negative impact on network performance. Additionally, it can be observed that when the separation numbers are set to 7, 3, 3, and 4, the network achieves optimal performance across four datasets. Therefore, in subsequent experiments, the separation numbers for the RJ1, RJ2, RJ3, and RJ4 datasets are set to 7, 3, 3, and 4, respectively.

Subsequently, without altering other structures of BSRNet, the authors separately removed the SepConv and APAConv convolutions from the Stage Layer to compare the performance changes of the network across four datasets. The results of these metrics are presented in Fig. 12, with detailed numerical data provided in Appendix A Table A. 1, Table A. 2, Table A. 3 and Table A. 4. Furthermore, to validate the substitutability and superiority of SepConv convolution over extra data preprocessing methods, the authors conducted denoising with Wiener filtering and feature selection using the classical Pearson correlation coefficient method after removing the SepConv convolution, simulating the scenario of employing extra preprocessing techniques.

It can be observed that the ablation study results across the four datasets exhibit similar trends. We discuss and analyze the results using the RJ1 dataset as an example. Specifically, on the RJ1 dataset, the recognition accuracy of BSRNet was 66.93 %. When the SepConv convolution was removed and only the APAConv convolution was active, the recognition accuracy dropped to 53.26 %. Conversely, when the APAConv convolution was removed, leaving only the SepConv convolution active, the recognition accuracy further decreased to 43.47 %. Similar downward trends were noted in other metrics as well. Additionally, after removing the SepConv convolution and applying extra preprocessing methods, the recognition accuracy improved from 53.26 % to 58.66 %, yet it did not reach the performance level attained with the SepConv convolution.

When the SepConv convolution is removed and only the APAConv convolution is active, the neural network effectively leverages the diverse local sound patterns captured by different convolution kernels, performing adaptive nonlinear mapping to enhance the model's ability to abstractly represent these patterns. However, it lacks the autonomy to capture core features, resulting in an inability to effectively separate useful information from redundant or noisy data during feature extraction. This ultimately limits the accuracy of recognizing broiler chicken sound signal types. In this scenario, although the use of extra preprocessing methods leads to an improvement in the model's recognition performance, it still does not achieve the accuracy in sound signal recognition attained when SepConv convolution is used. This demonstrates that while extra preprocessing methods can enhance the network's performance to some extent, the improvement is limited. It further indicates that, when using SepConv convolution as the feature extraction layer in the network, there is no need for overly complex or cumbersome preprocessing steps (such as filtering, denoising, or feature

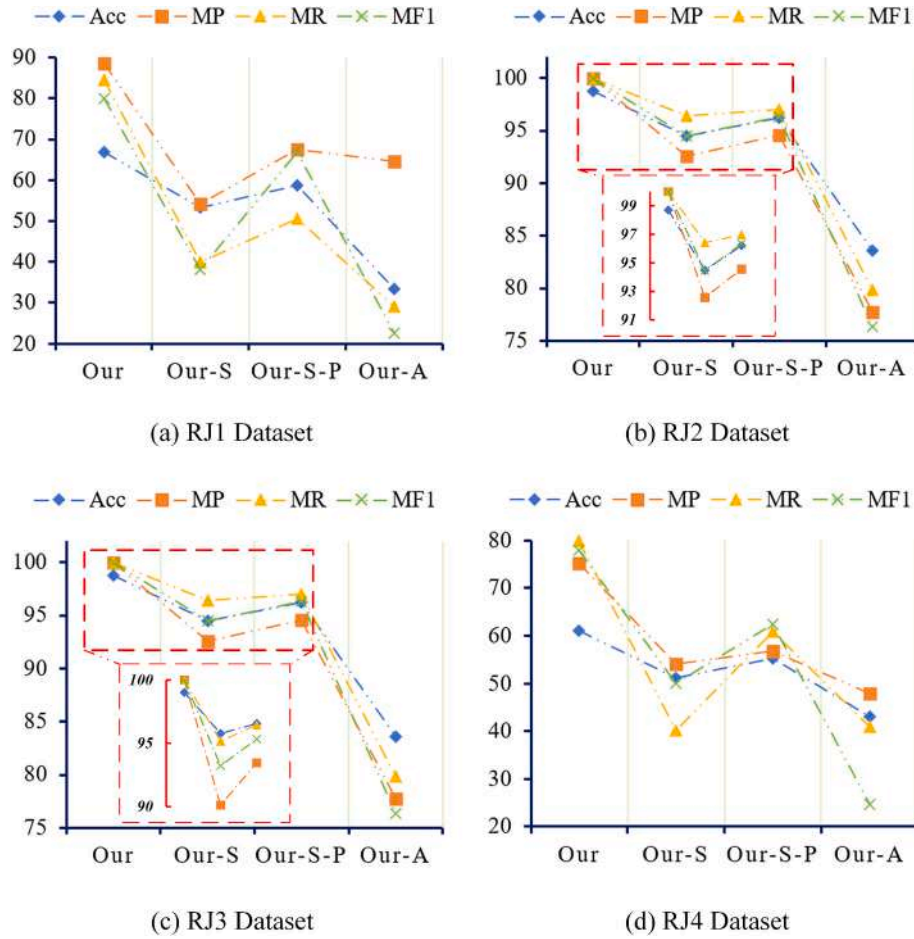


Fig. 12. Ablation study results on four datasets. 'Our-S' and 'Our-A' represent the experimental results after removing the SepConv and APAConv convolutions, respectively. 'Our-S-P' represents the experimental results after removing the SepConv convolution and applying extra preprocessing (Wiener filtering and Pearson correlation) instead.

selection) on the raw audio signals in order to achieve competitive recognition performance.

Specifically, typical extra preprocessing methods consist of two steps: first, the Wiener filtering algorithm is applied to suppress noise in the sound signals; next, the Pearson correlation coefficient is used to select and analyze the importance of each dimension of the acoustic features. Features with lower importance are considered redundant or irrelevant and are discarded, which helps ensure that the acoustic features input into the model are core and relevant. However, the relationship between acoustic features and sound signal categories is complex and nonlinear. Although the extra preprocessing methods can filter out noise and redundant features, they also remove the latent nonlinear information embedded in those features. This hinders the network's ability to understand and establish the complex nonlinear relationship between inputs and outputs, ultimately limiting the accuracy of sound signal recognition. As discussed by the authors in [Section 3.4.1](#), the effectiveness and applicability of such methods warrant further investigation.

When only the SepConv convolution is active and the APAConv convolution is removed, the network can autonomously identify useful core features within each stage, enhancing sensitivity to important features. However, the network is unable to fully represent the complex sound mapping patterns contained in useful features as captured by different convolution kernels. This leads to a failure to fully learn the potential correlations among features, weakening the network's deep understanding of the complex relationships between inputs and outputs, and adversely affecting the network's ability to accurately identify

sound signal categories.

Furthermore, the performance degradation of the neural network when the APAConv convolution is removed is significantly greater than when the SepConv convolution is removed. This suggests that, although the network possesses the ability to autonomously capture core useful features and can effectively separate irrelevant or noisy features, it fails to adaptively perform nonlinear mapping based on the specificity of core features captured by multiple convolution kernels. The network lacks flexibility in abstracting local broiler chicken sounds with different representational meanings, which makes it difficult to ensure the richness and precision of the extracted core features. As a result, this leads to a more substantial overall decrease in recognition accuracy.

In summary, the ablation study results validate the effectiveness of the two novel convolutions. The SepConv convolution focuses on the autonomous capture of useful core features, while the APAConv convolution enhances the feature expression ability for various local broiler chicken sound signal patterns extracted by multiple convolution kernels. The combination of both not only enhances the network's ability to express acoustic features but also improves the effectiveness and richness of core feature extraction, effectively increasing the accuracy of broiler chicken health monitoring.

3.3. Comparative experiments

This subsection compares BSRNet with eight other methods to validate its superior performance and advanced nature. Specifically, the authors trained eight sound signal recognition models using the four

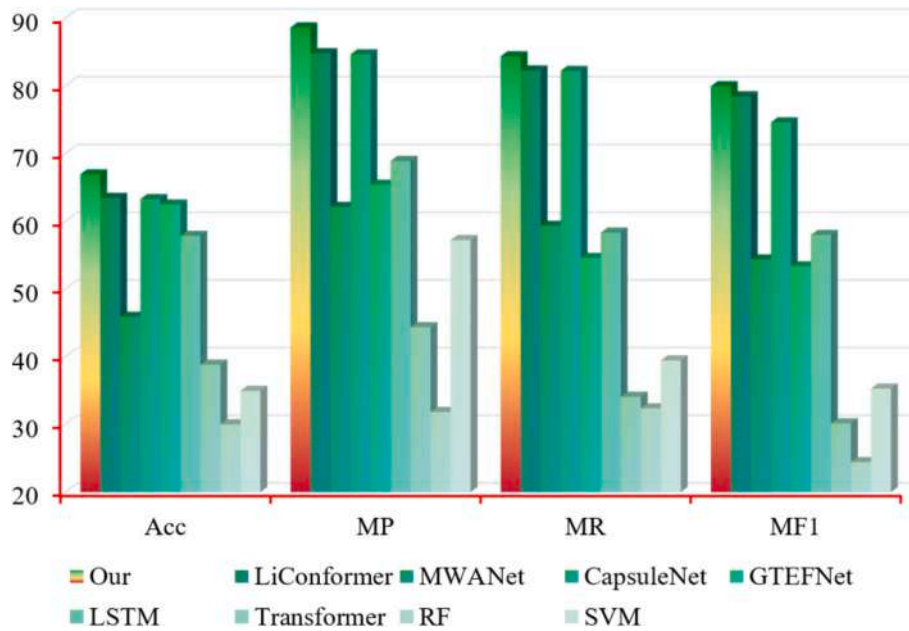


Fig. 13. Comparison of experimental results on the RJ1 dataset with other methods.

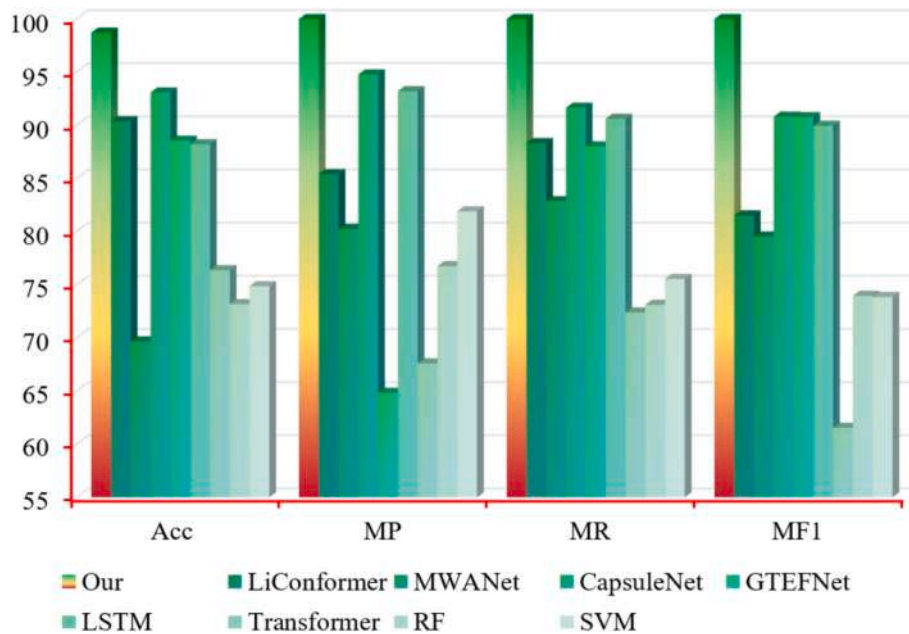


Fig. 14. Comparison of experimental results on the RJ2 dataset with other methods.

datasets constructed in Section 2.1.2, including two machine learning models (SVM (Sun et al., 2023a), Random Forest (Sun et al., 2023b)) and six CNN models used for signal recognition (LiConformer (Yan et al., 2024), MWANet (Wang et al., 2023), CapsuleNet (Chen et al., 2022), GTEFNet (Jia et al., 2023), LSTM (Yang et al., 2024), Transformer (Liu et al., 2024)). To ensure the fairness of the experiments and maximize the performance of these methods, while also facilitating the replication of results by other researchers, the settings for the hyperparameters were kept consistent with those in the original papers of the comparative methods.

The comparison experimental results across the four datasets are shown in Fig. 13, Fig. 14, Fig. 15 and Fig. 16, with detailed numerical data provided in Appendix A Table A. 5, Table A. 6, Table A. 7 and Table A. 8. The results demonstrate that the proposed BSRNet achieves

competitive performance across all four evaluation metrics. Furthermore, almost all methods perform better on the RJ3 dataset compared to the other datasets. Therefore, the authors will focus on the RJ3 dataset for further discussion and analysis. Specifically, on the RJ3 dataset, BSRNet achieved superior performance with an Acc of 99.02 %, outperforming other methods by 9.05 % to 25.09 %. This indicates that BSRNet delivers outstanding performance in broiler chicken sound type recognition. In terms of MP, MR, and MF1, BSRNet achieved 100 % performance, surpassing other methods by 4.63 % to 38.46 %. This demonstrates that BSRNet offers high reliability in recognizing positive instances of broiler chicken sound signals while effectively reducing false positives. Additionally, BSRNet shows significant advantages in coverage for broiler chicken sound types, effectively capturing positive samples and significantly reducing missed detections. It also achieves a

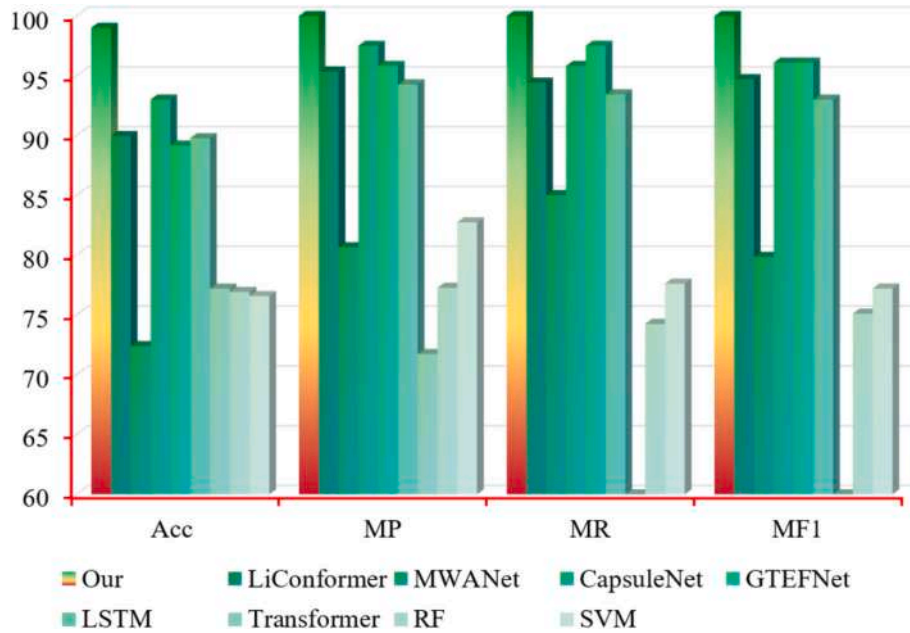


Fig. 15. Comparison of experimental results on the RJ3 dataset with other methods.

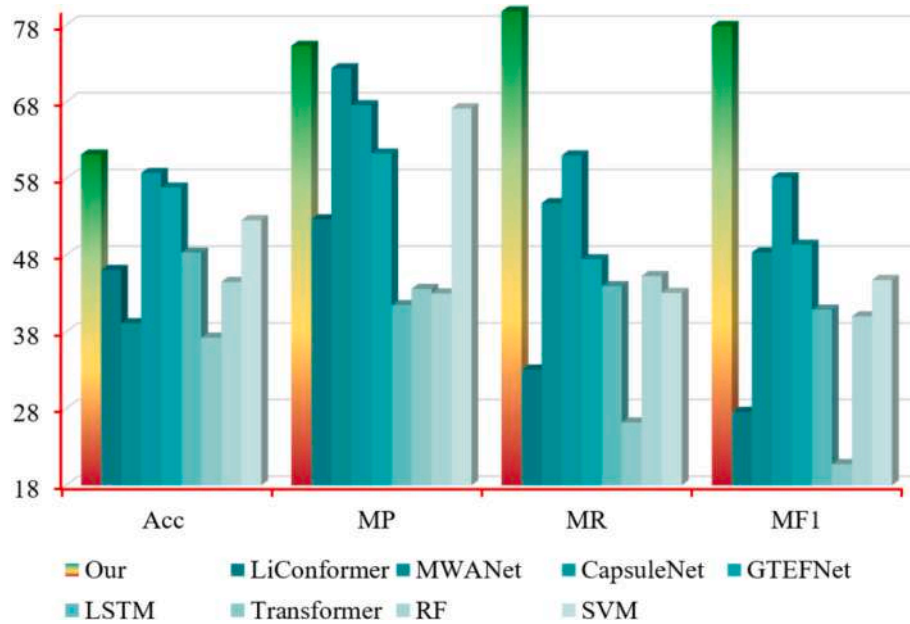


Fig. 16. Comparison of experimental results on the RJ4 dataset with other methods.

better balance between precision and recall, resulting in higher overall classification performance, which enhances its practical application in real-world scenarios. In fact, on the other three datasets, BSRNet exhibits similar advantages in sound signal recognition performance. The above experimental results validate the superiority and advancement of the BSRNet proposed in this paper.

To more accurately evaluate the performance of various methods in the broiler chicken sound signal classification task, the authors used confusion matrices for a more comprehensive analysis, with experimental results shown in Fig. 18 and Fig. 17. The results indicate that the proposed BSRNet achieves competitive performance across all broiler chicken sound signal classifications. In contrast, other methods fail to achieve accurate recognition across all categories and exhibit varying degrees of misclassification. This further validates the advantages of

BSRNet in classification accuracy and robustness. To assess the dynamic performance of the proposed method and other methods under different thresholds, particularly in terms of overall performance under class imbalance conditions, the authors employed ROC curves and the AUC metric for a deeper and more comprehensive evaluation. The experimental results are presented in Fig. 19. The results demonstrate that BSRNet's ROC curves for C0, C1, C2, C3, C4, C5, C6, and C7 (8 sound signal categories) are close to the top left corner, with corresponding AUC values reaching 1.00. This indicates that the model exhibits strong overall discriminative ability across all sound signal categories, significantly outperforming other methods. In contrast, the curves of the comparison methods show considerable fluctuations and noticeable jitter, indicating weaker discrimination capabilities between different categories. For example, CapsuleNet's AUC for C6 is only 0.82, and

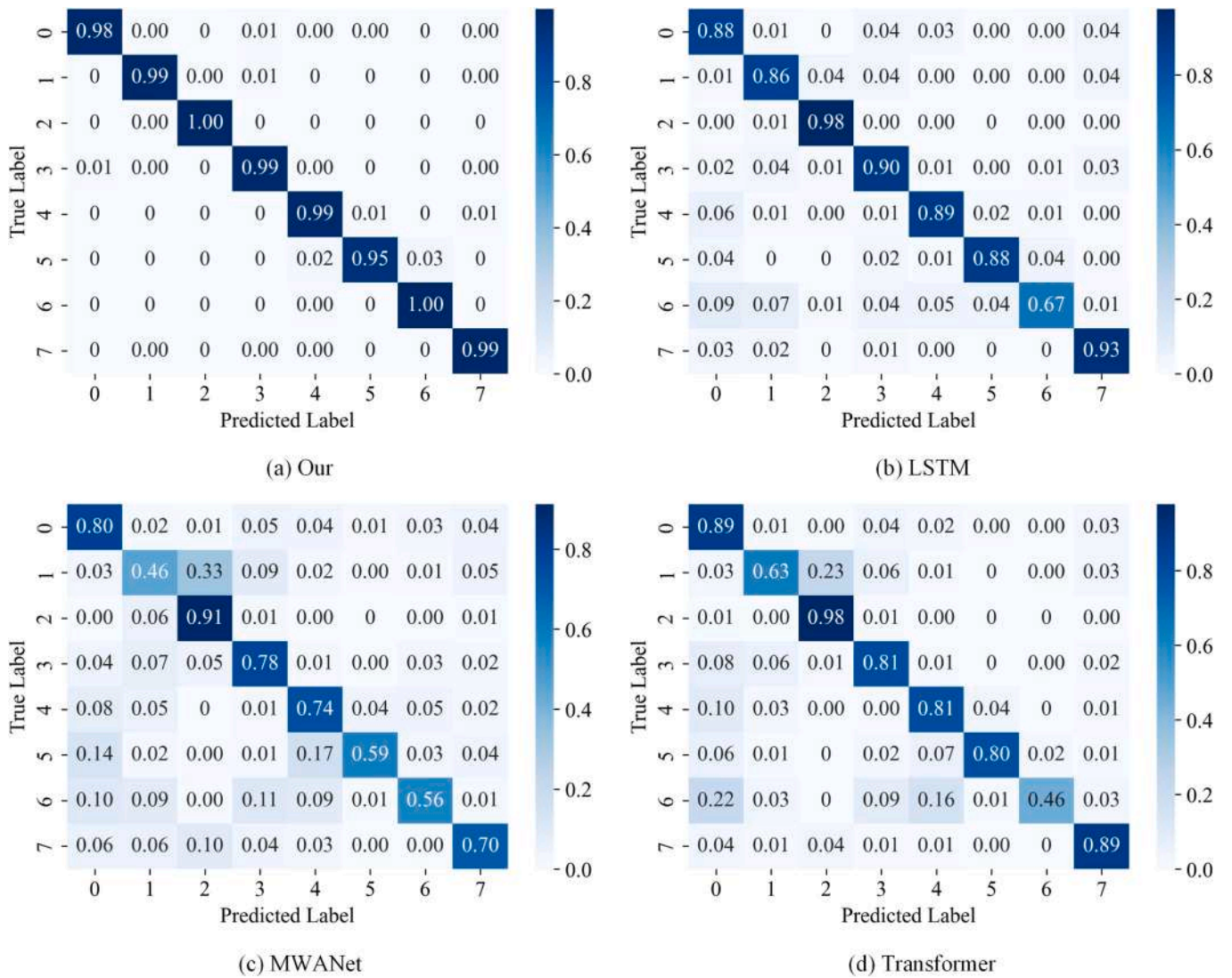


Fig. 17. Confusion matrix (I) for broiler chicken sound signal recognition results compared to recent methods.

GTEF's AUC for C4 is 0.85. Additionally, BSRNet's MAAUC value is 1.00, confirming that the proposed method demonstrates stable overall classification performance under class imbalance conditions, outperforming other methods. For instance, it is 10 % higher than GTEF (MAAUC = 0.90) and 11 % higher than CapsuleNet (MAAUC = 0.89). These experimental results further validate the superiority of the proposed method in terms of overall classification capability.

In practical application scenarios, in addition to the accuracy of target recognition, model parameters and inference time are also critical metrics. To evaluate the practicality of the proposed method, the authors summarize the training time, inference time, and model parameter count of the proposed method and several other methods (with recognition accuracy for sound signals above 85 %), as shown in Table 6. The results demonstrate that, compared to other methods, the proposed method also achieves competitive performance in terms of training time, inference time, and model parameter count, with certain advantages over other methods, including lightweight networks such as LiConformer and GTEFNet.

Specifically, LiConformer is a lightweight model that uses a broadcast self-attention mechanism to capture fine-grained features of the signal globally. However, attention mechanisms typically rely on weights generated by external sub-networks to compute the relevance of each feature, which consumes additional training resources and inference time. GTEFNet, on the other hand, integrates a denoising strategy

into the network, which effectively reduces noise interference and improves signal distinguishability. However, since the denoising strategy is similar to traditional preprocessing methods, its integration into the network introduces additional time consumption from the preprocessing stage into the model's training and inference phases, increasing both training and inference time. Although the LSTM method performs the best in terms of model parameters, the proposed method is the second best. However, LSTM's performance in sound signal recognition is limited and fails to meet the high accuracy and robustness required for practical applications. In contrast, the proposed method, starting from the core component of convolution operations in neural networks, introduces two novel convolutions and a DFF structure, achieving high-precision sound signal recognition without relying on external sub-networks or similar preprocessing methods. This effectively avoids the additional resource consumption from training sub-networks and integrating preprocessing methods. In summary, the proposed method not only demonstrates high recognition performance but also offers advantages in computational resources and time efficiency, meeting the requirements of practical applications. This further validates its practicality in real-world scenarios.

3.4. Interpretability analysis

This subsection provides a comprehensive interpretability analysis of

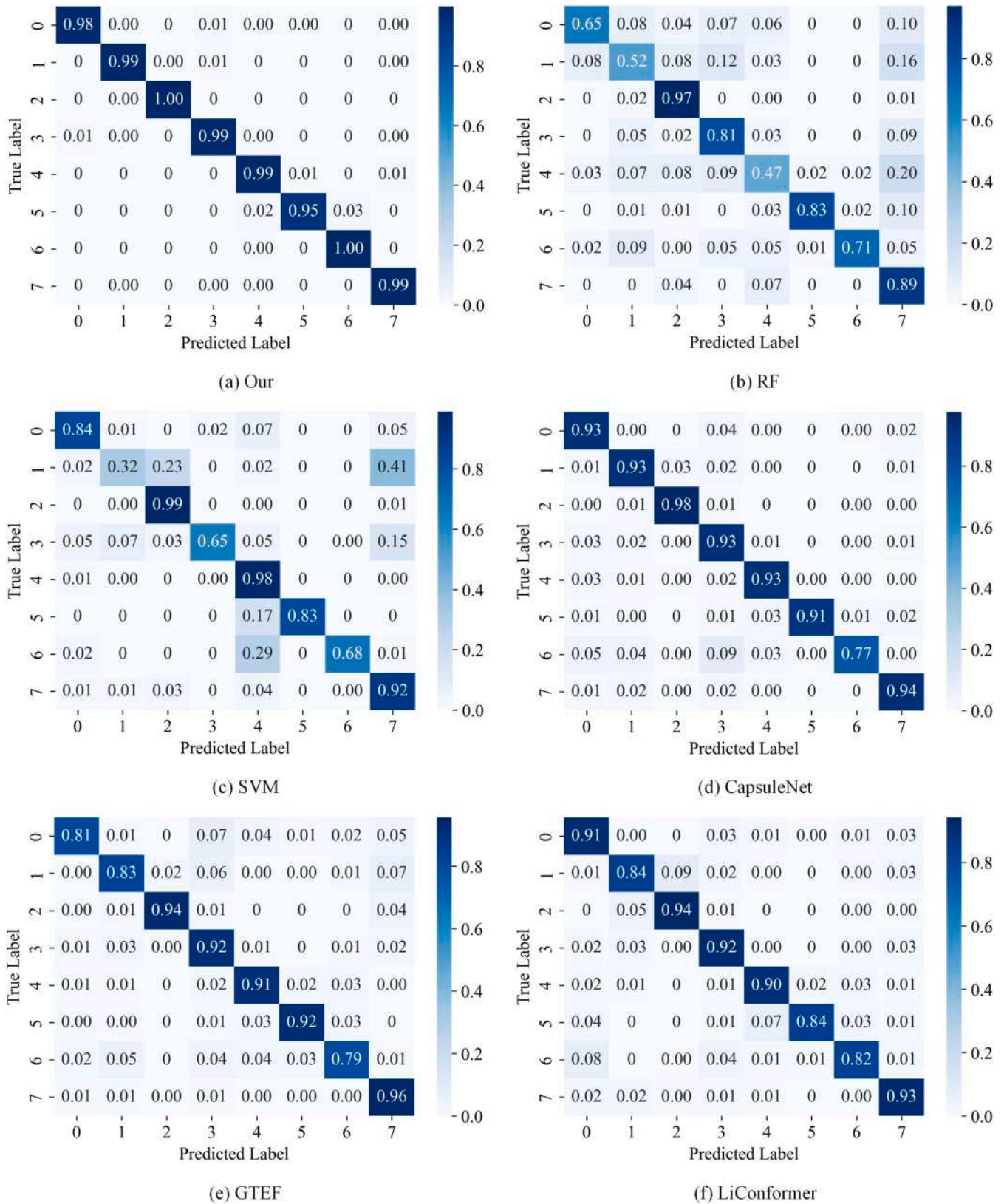


Fig. 18. Confusion matrix (II) for broiler chicken sound signal recognition results compared to recent methods.

BSRNet using Grad-CAM++ technology, T-SNE technique, and feature extraction visualization.

3.4.1. Interpretability analysis 1

The authors randomly selected one sample of each of the 8 types of broiler vocal signals from the four datasets and used the Grad-CAM++ technique to establish the attention visualization relationship between

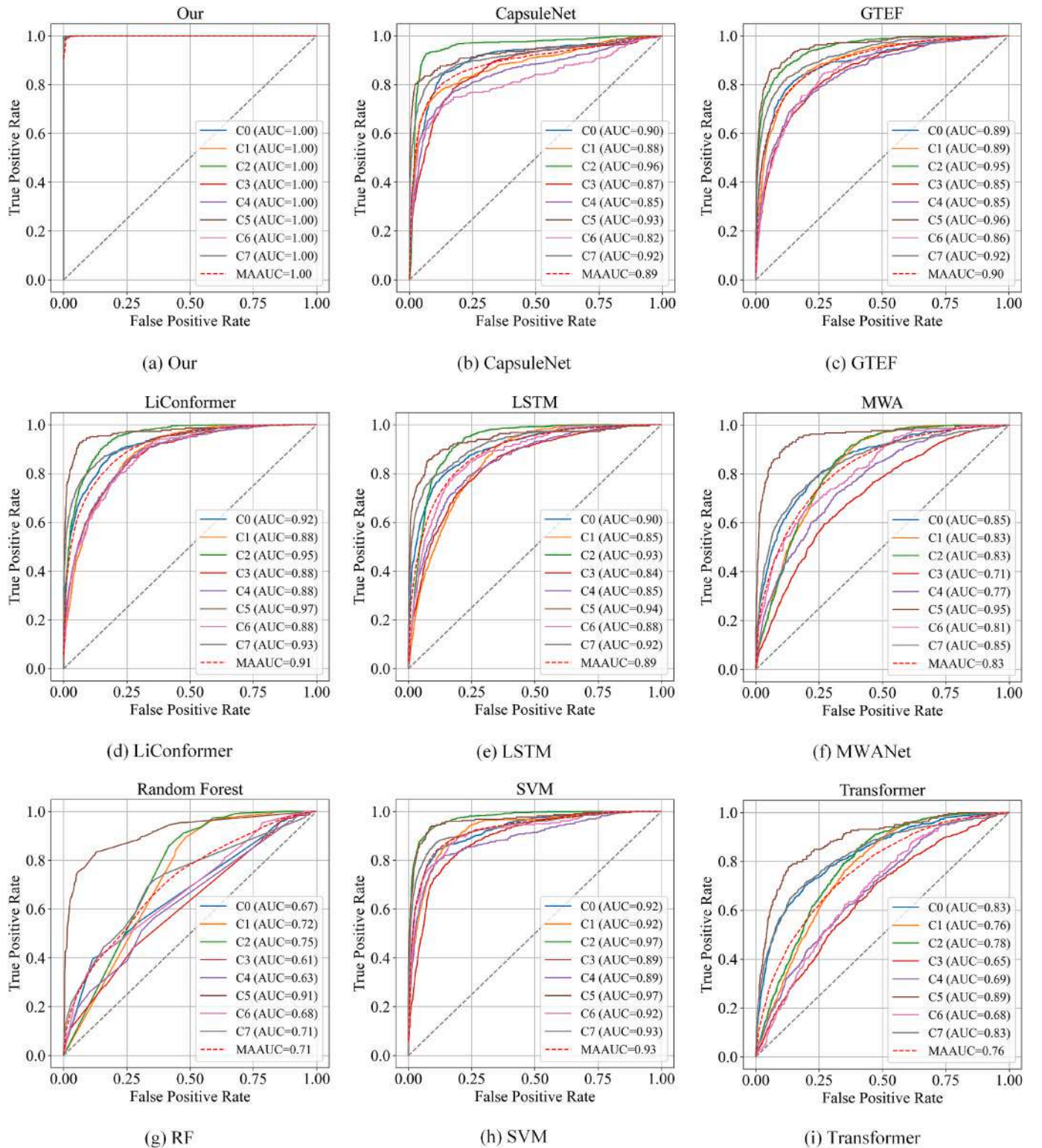


Fig. 19. ROC curves and AUC metrics for broiler chicken sound signal recognition compared to recent methods. 'MAAUC' represents Macro AUC. C0 to C7 indicates the trend of the ROC curve and the corresponding AUC indicator for the recognition of 8 types of sound signals by the model.

BSRNet and the broiler vocal signal features, analyzing the model's attention to the input features. The experimental results are shown in Fig. 20 and Fig. 21, where the horizontal axis represents the dimensionality of the acoustic features. Feature sample points with darker colors indicate that the neural network assigned higher attention weights to these features during the classification decision process, meaning that these features have a higher contribution and impact on

the network's recognition results. The red dashed line represents the numerical results of the network's attention to different feature sample points.

The experimental results show that on the four datasets, only some feature sample points are darker in color, indicating they play a crucial role in the network's classification decisions, while most sample points are lighter, indicating they contribute less. Furthermore, within each

Table 6

Comparison of model complexity. 'T-Time' and 'I-Time' refer to training time and inference time, respectively. 'Params' indicates the model parameter count. Bold indicates optimal performance, and italic indicates suboptimal performance.

Methods	Acc (%)	T-Time (s)	I-Time (s)	Params (M)
Our	99.02	901.351	1.13	2.78
LiConformer	89.97	989.465	3.49	6.2
CapsuleNet	93.02	1066.233	4.06	78.39
GTEFNet	89.17	1127.476	4.61	3.01
LSTM	89.78	983.524	1.22	2.51

dataset, the color distribution of sample points from different sound signal types varies. This confirms that BSRNet has the autonomy to identify core features in different sound signal type feature sample points and dynamically assigns high attention to these parts, while dynamically diminishing attention to features with lower contributions. Combined with the results of the ablation study in Section 3.2, this autonomy helps enhance the model's recognition performance. It further proves that not all acoustic features contribute significantly to the recognition results. It is noteworthy that while the network's attention to most low-contribution features is lower, it is not zero, suggesting that even features with low contributions might play a role under specific circumstances. Therefore, the applicability of traditional feature selection methods that directly eliminate low-contribution features requires further exploration.

Furthermore, compared to the samples in the RJ1 and RJ4 datasets, the network exhibits more precise recognition and capture of core features in samples from the RJ2 and RJ3 datasets. This indicates that, compared to the Fbank and MFCC algorithms, the Spectrogram and Mel Spectrogram algorithms provide a more accurate and in-depth exploration of the complex nonlinear relationship between physical information and sound signal types, offering more valuable information to the network. As demonstrated by the experimental results in Section 3.3, the network's recognition accuracy on the RJ2 and RJ3 datasets surpasses that on the RJ1 and RJ4 datasets. This validates the importance and necessity of extracting physical information from sound signals to introduce prior knowledge into the subsequent CNN processing.

3.4.2. Interpretability analysis 2

Subsequently, to analyze the feature extraction capabilities at different stages of BSRNet, the authors used the T-SNE technique to visually analyze the feature extraction capabilities of different blocks in BSRNet, as well as the feature distribution in the Global Average Pooling (GAP) layer and the FC Block. Here, we discuss the results using the RJ3 dataset as an example, with experimental results shown in Fig. 22. The results indicate that as the input progresses from Block 1 to Block 4, the features become more organized. Although the feature distribution in Block 1 is relatively chaotic, as the network deepens, clearer clustering trends begin to emerge. In Block 4, the feature points of different categories start to distinctly separate, indicating that the network is capable of effectively distinguishing between broiler chicken sound types at this stage. In the visualizations of the GAP layer and FC Block, the features are further compressed and clustered. Particularly in the FC Block, feature points of different categories are clearly separated, forming independent feature clusters. This demonstrates that after mapping through the FC Block, the model has established a mapping relationship between features and broiler chicken sound types, enabling more accurate classification of samples.

In summary, BSRNet first progressively extracts and refines features through four blocks, then effectively compresses and aggregates the features in the GAP layer, and finally enhances the discriminative power of the features and achieves efficient classification through the FC Block. This structural design demonstrates a strong performance advantage when BSRNet handles complex sound signal classification tasks, further validating the rationality and effectiveness of the BSRNet architecture.

3.4.3. Interpretability analysis 3

Although the T-SNE technique has been used to analyze the capability of BSRNet at a macroscopic level in handling broiler chicken sound features through its various stages, the microscopic details of how each layer of the model captures and transforms features, and how this promotes the final classification decision process, remain unclear. To address this, we visualized the output features at different stages of the model, including the four Block stages and the final FC Block. The experimental results are shown in Fig. 23, Fig. 24, Fig. 25, Fig. 26, Fig. 27 and Fig. 28. In these feature maps, the vertical and horizontal directions represent the feature dimensions and channel dimensions, respectively, with the orientation described starting from the bottom left corner of the feature map.

From the visualization results in Fig. 23, it can be observed that features with similar or closely related representational meanings exhibit texture variations with the same color (textural characteristics), while features with significantly different or divergent meanings show edge-like changes with noticeable color shifts (edge characteristics). These two types of variations appear in both the feature and channel dimensions, initially forming horizontal and vertical strip-like textures (referred to as horizontal and vertical textures), typically accompanied by some degree of edge characteristics. Specifically, the horizontal texture reveals the variation patterns of features' abstract representations across different channels, reflecting the interrelationships and trends of features across multiple channels, while the vertical texture represents the abstract representation of features within the same channel. This phenomenon suggests that, at the Block 1 stage, BSRNet effectively extracted edge and texture information embedded in the input data, preliminarily learning the complex interactions between features and across channels, providing essential support for higher-level feature learning and classification. Compared to Block 1, the feature maps extracted at Block 2 exhibit more pronounced and refined horizontal and vertical textures, capturing more semantically meaningful detailed features and transforming them into higher-level feature representations. This aids the model in understanding the underlying patterns in sound signals.

Compared to the first two stages, Block 3 and Block 4 perform deeper and more comprehensive extraction and refinement of features, enabling the model to capture more abstract information. These blocks re-represent the characteristic patterns of physical information in broiler chicken sound signals, the potential relationships between physical information, and the complex nonlinear relationships with sound signal categories, resulting in more advanced, abstract, and comprehensive output feature maps. Additionally, compared to Block 3, Block 4 demonstrates a stronger ability to refine and abstract the acoustic features. Building upon Block 3, it further extracts features at a deeper and more refined level. The distribution of horizontal and vertical textures in Block 4 is more uniform, detailed, and regular, indicating that, at the Block 4 stage, the network has a more thorough understanding of the correlations between features across different channels and the potential relationships between features. The captured information is more abstract, comprehensive, and refined. These advanced features effectively reflect the latent patterns within the signal, providing richer and more suitable feature representations for broiler chicken health monitoring.

The FC Block, through the fully connected layer, transforms the sound features extracted at each stage into globally discriminative feature representations and maps them into the decision space, assisting the model in making the final decision. The output feature map is shown in Fig. 27. Each small square in the figure represents the model's score for a sample belonging to a certain category, with the eight small squares corresponding to the unnormalized scores for the eight categories. Different color arrangements indicate which type of sound signal the sample is associated with. It can be observed that there are eight possible color combinations, indicating that the model is able to accurately distinguish the eight types of sound signals. In fact, the visualization result shown in Fig. 27 remains abstract. To further demonstrate how

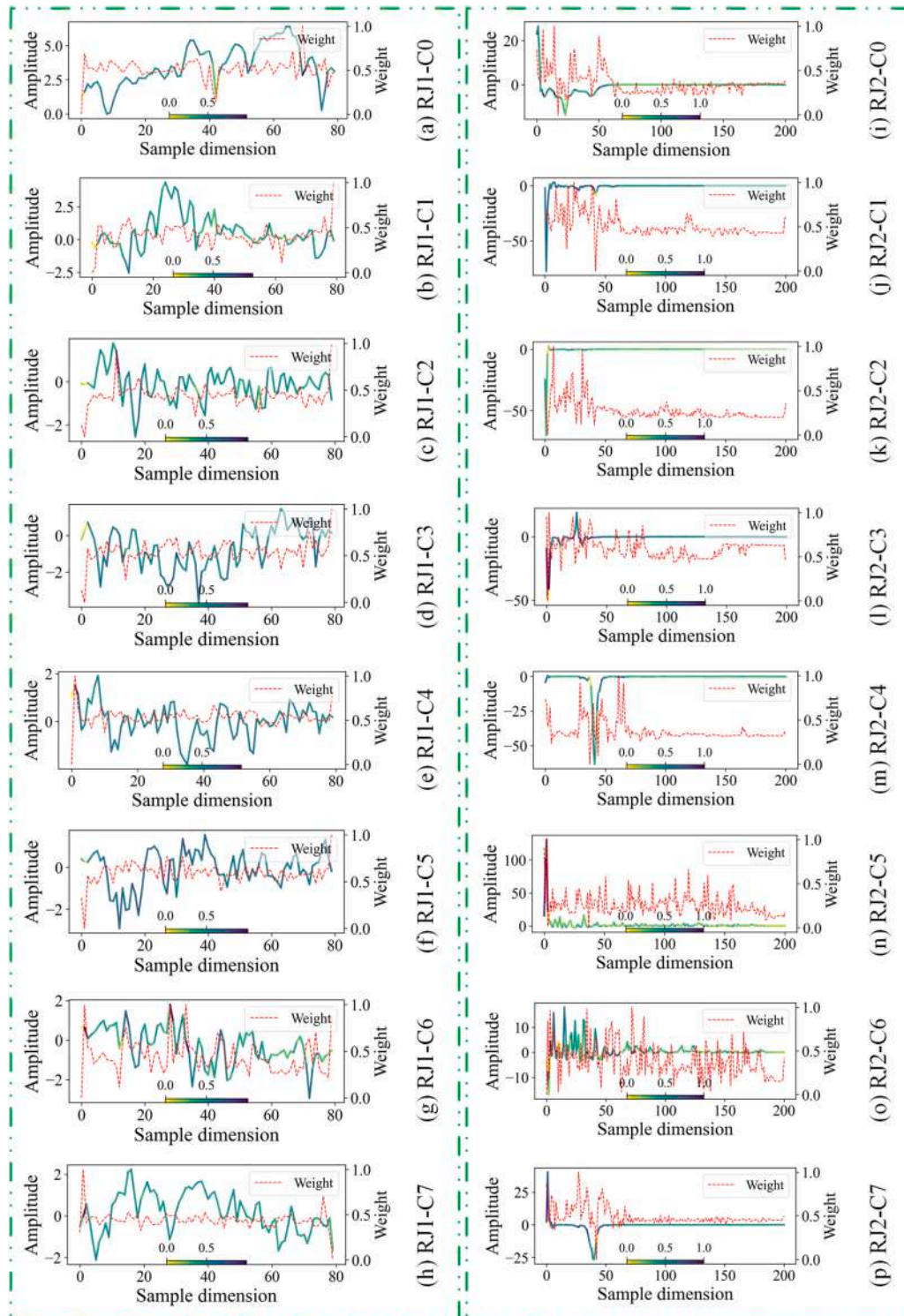


Fig. 20. The network's attention relationship with input feature samples on the RJ1 dataset (acoustic feature dimension of 80) and the RJ2 dataset (acoustic feature dimension of 201). Where C0 ~ C7 represent the acoustic feature samples corresponding to the 8 types of vocal signals.

the FC Block maps the sound signals, the authors applied Softmax to the feature map of FC Block, which makes it clearer to show the recognition and mapping of specific categories, as shown in Fig. 28. After applying Softmax, the mapping relationship between the sample inputs and the categories becomes clearer. The index of the brown square indicates the category of the sample. This shows that, during the process of feature extraction and abstraction in the first four blocks, the model has achieved a progressive learning process, from low-level features to high-

level semantic features. Additionally, through the deep processing and reconstruction of features, the model uncovers complex nonlinear patterns and category-related information hidden within the features. Ultimately, these progressively extracted high-dimensional features are fused and mapped in the fully connected layer, providing comprehensive and discriminative feature representations for the decision space, which effectively improves the model's classification performance and interpretability. These experimental results once again validate the

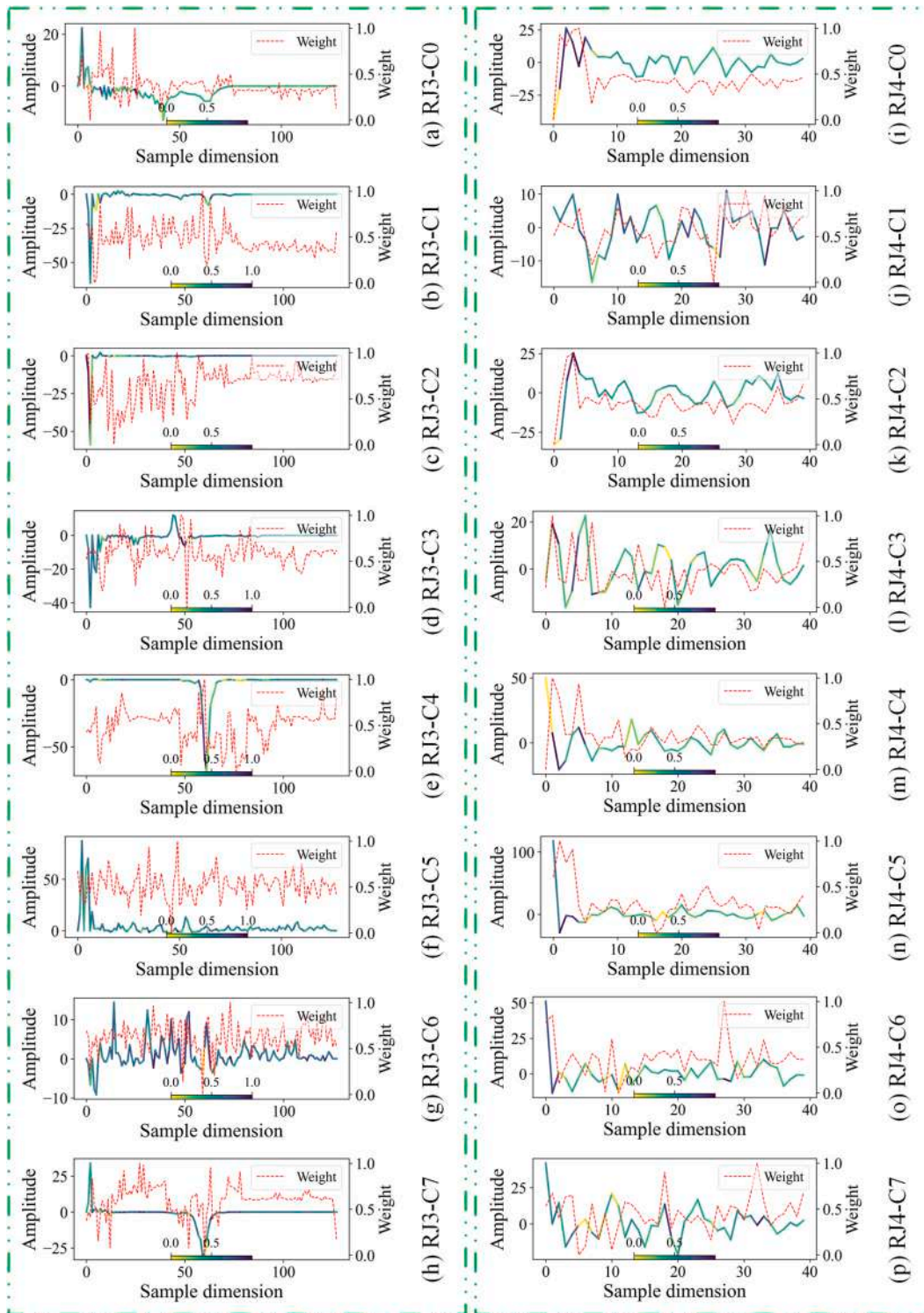


Fig. 21. The network's attention relationship with input feature samples on the RJ3 dataset (acoustic feature dimension of 128) and the RJ4 dataset (acoustic feature dimension of 40). Where C0 ~ C7 represent the acoustic feature samples corresponding to the 8 types of vocal signals.

rationality and effectiveness of the BSRNet structure.

3.5. Discussion of practical application strategies and limitations of the method

This section discusses the practical application strategies, limitations, and future research directions of the method presented in this paper. Specifically, in practical scenarios of broiler health monitoring,

requirements such as low computational cost and high accuracy are often necessary. Based on the broiler's vocal signals, this paper proposes a novel neural network for broiler chicken sound signal recognition (BSRNet) for health monitoring. The experimental results presented earlier in the paper have demonstrated that BSRNet can rapidly assess broiler health status by analyzing the vocalizations emitted by the broilers, without the need for additional preprocessing of the original sound signals. This method satisfies the practical demands of low

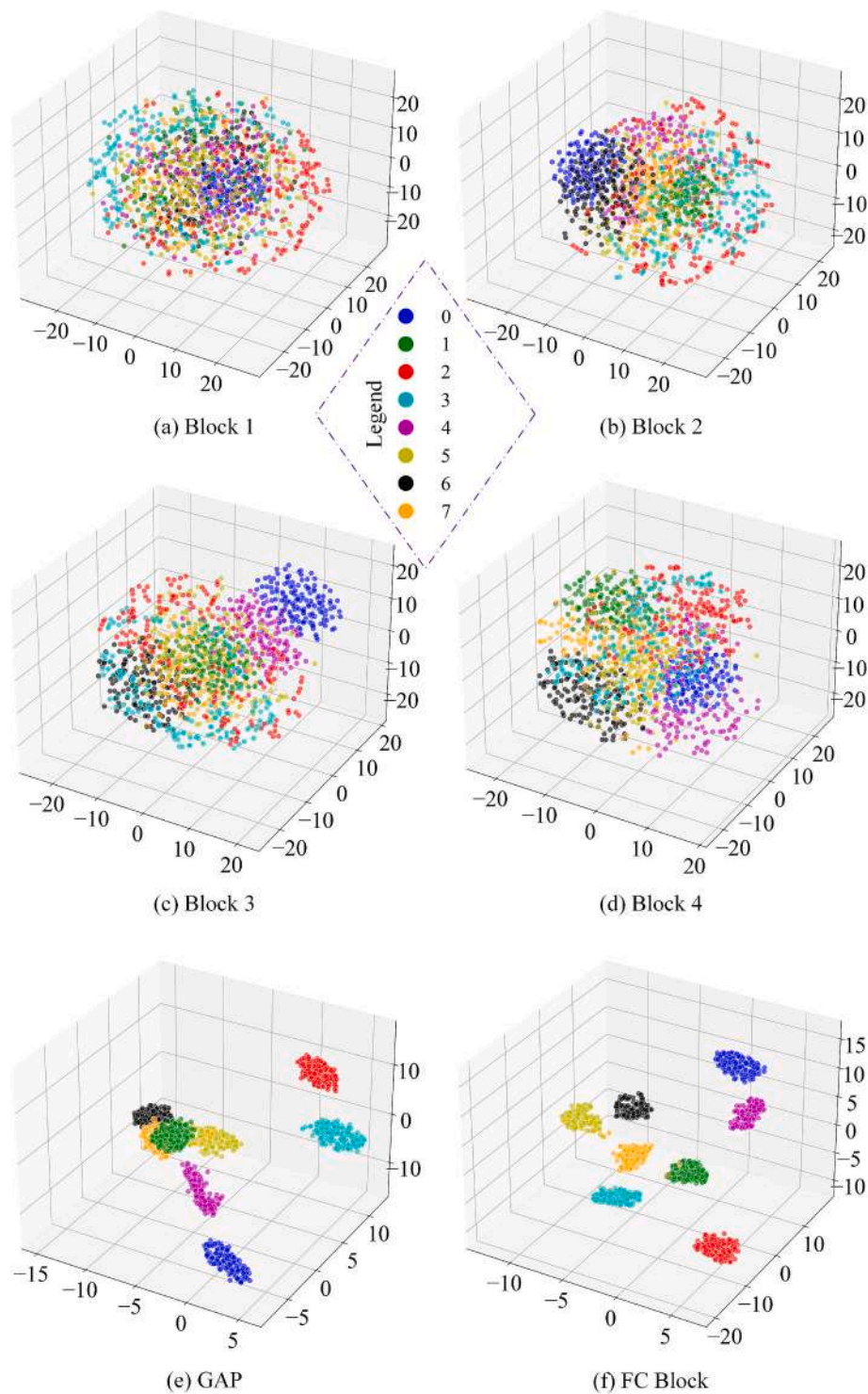


Fig. 22. Visualizing the features of each stage of BSRNet using T-SNE.

computational power and high accuracy, providing a core methodological foundation for non-contact, automated health monitoring of broilers in scenarios with limited computational resources. Additionally, with the assistance of poultry farm managers and veterinary husbandry experts, data collection and annotation were completed following the experimental steps outlined in [Section 2.1](#), ensuring a strong correlation between each recorded sound signal and the health status of the broiler chickens. This is crucial for training a high-accuracy sound signal recognition model and is a core and necessary condition for applying the model in complex farming environments.

In group-based health monitoring scenarios, where multiple broilers are housed together, we do not directly use BSRNet to perform complex sound source separation and classification of the mixed sounds of multiple broilers. Instead, we treat it as an efficient and scalable abnormal acoustic event monitoring and classification system. In this paper, we discuss the practical application strategy using the farming environment described as an example.

First, based on the description in [Section 2.1.1](#), the entire greenhouse farming structure consists of multiple sub-areas, each measuring 2 m in both length and width. Our strategy takes full advantage of this layout.



Fig. 23. Visualization of Block 1 Output Features.

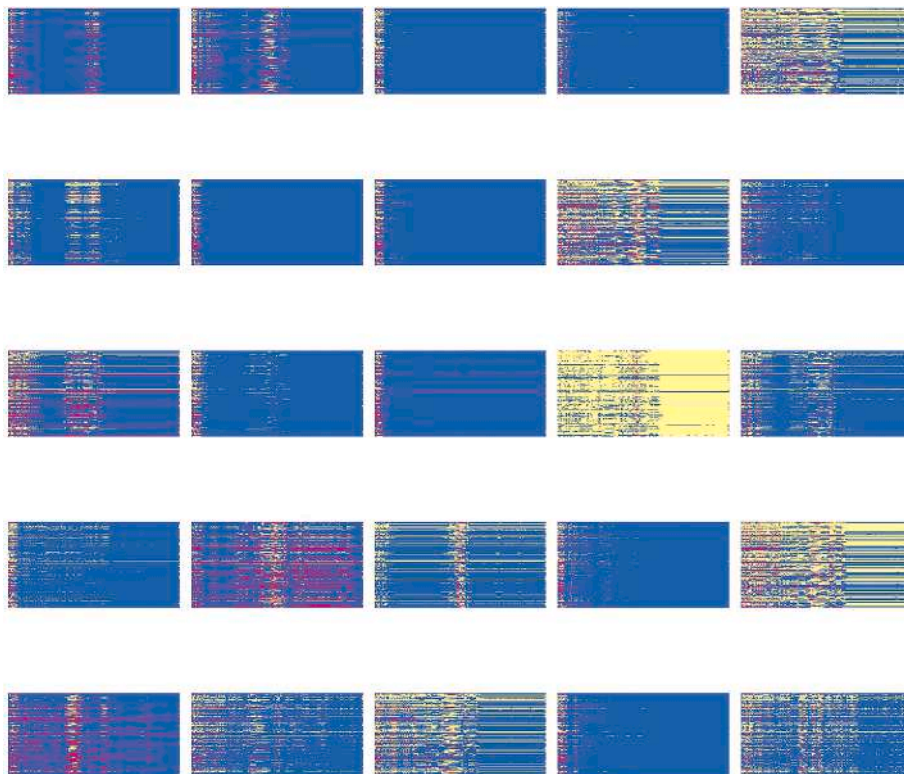


Fig. 24. Visualization of Block 2 Output Features.



Fig. 25. Visualization of Block 3 Output Features.

Specifically, in each sub-area, a basic monitoring system is deployed, consisting of a microphone, a computational unit equipped with a pre-trained BSRNet model, a simple counting algorithm, and a wireless transmission module. It is important to note that the microphone's pickup range is designed to cover only the designated sub-area and does not overlap with the adjacent areas, ensuring that the microphone clearly captures the sounds within its area while minimizing interference from neighboring regions. Moreover, existing studies indicate that in everyday environments, healthy broilers are relatively quiet most of the time (Sun et al., 2024). This means that abnormal sounds emitted by a few sick broilers will often stand out as isolated or sudden acoustic events, which are then captured by the microphone deployed in the corresponding area. This facilitates the subsequent classification and statistical analysis of the acoustic events by the BSRNet.

Next, once the basic monitoring system is activated in each sub-area, the microphone continuously captures sound. These audio clips are then input in real-time into the corresponding BSRNet. Thanks to BSRNet's powerful end-to-end recognition capabilities, the model can efficiently and accurately classify various sound events directly from the raw audio, especially key sounds related to diseases, such as coughing and wheezing. Finally, the basic monitoring system uses the counting algorithm to independently tally the frequency of disease-related sounds (e.g., coughing, chirping) detected within the sub-area during a unit of time (e.g., per hour or per day). If the frequency exceeds a certain threshold or shows an abnormal rapid increase, it indicates that respiratory diseases may be spreading within the area. The system will automatically trigger an alert and transmit the warning information, frequency data, and location information to a remote receiver via the wireless transmission module, notifying veterinary husbandry experts to conduct timely professional inspections and interventions to prevent

large-scale outbreaks.

In summary, by deploying the basic monitoring system in each sub-area of the greenhouse as described, scalable group health monitoring can be achieved, providing a clear understanding of the dynamic health status of the broiler population. This practical application strategy is similar to recent methods for group broiler health monitoring (Tao et al., 2024; Sun et al., 2023a; Sun et al., 2024). However, compared to these methods, BSRNet proposed in this paper does not require complicated preprocessing such as filtering and denoising of the original audio. It can autonomously learn and focus on core pathological features in an end-to-end manner, resulting in higher recognition accuracy, robustness, and lower computational cost in complex real farming environments.

It is important to emphasize that the strategy based on BSRNet is highly versatile. In farming environments similar to those described in this paper, the deployment can follow the strategy outlined above. In farming environments with different layouts, even without sub-areas as described, management personnel only need to adhere to the core idea of zonal monitoring. Based on the actual pen layout, they can delineate monitoring units and adjust the strategy accordingly. For example, microphone models with appropriate pickup ranges, the selection of deployment locations, the definition of time units, and the standard values for abnormal sound frequency alerts of sick broilers need to be adjusted according to the specific environment. Moreover, regardless of whether the farming environment is consistent with the one described in this paper, the success of the entire monitoring system fundamentally depends on the sound signal recognition model. Therefore, BSRNet, with its end-to-end, high-precision, and low-cost characteristics, provides the key methodological foundation for the practical application strategies in this field, which represents the core value of this study.

To validate the effectiveness of the aforementioned practical

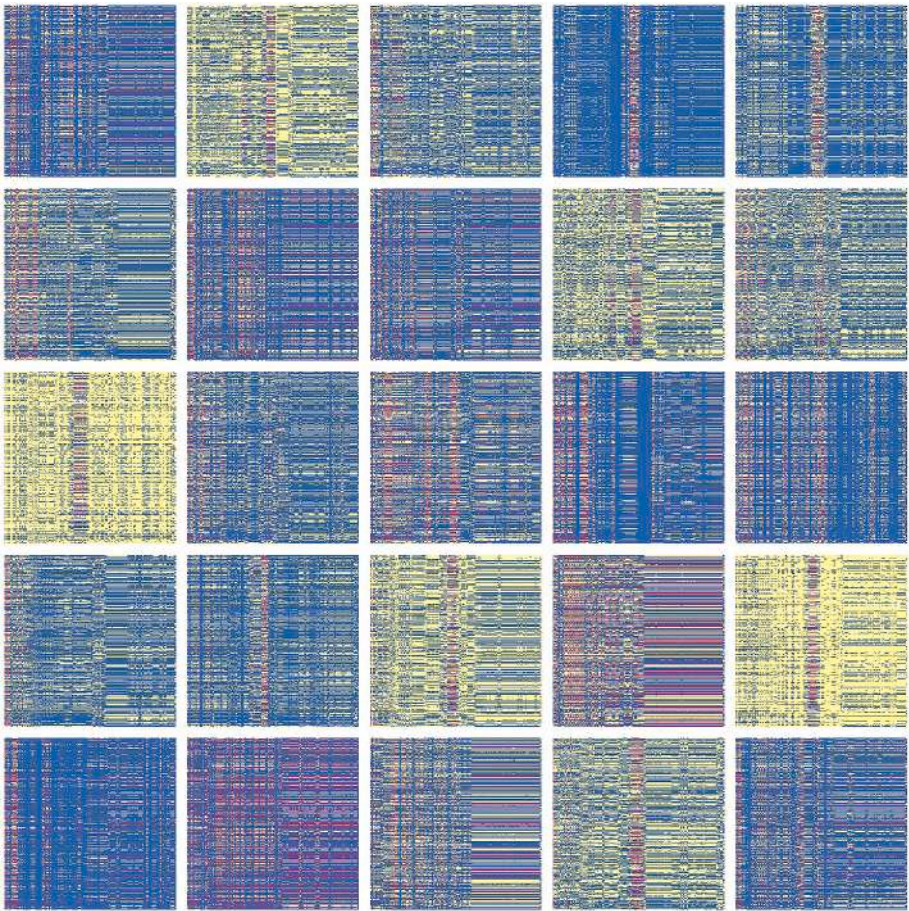


Fig. 26. Visualization of Block 4 Output Features.

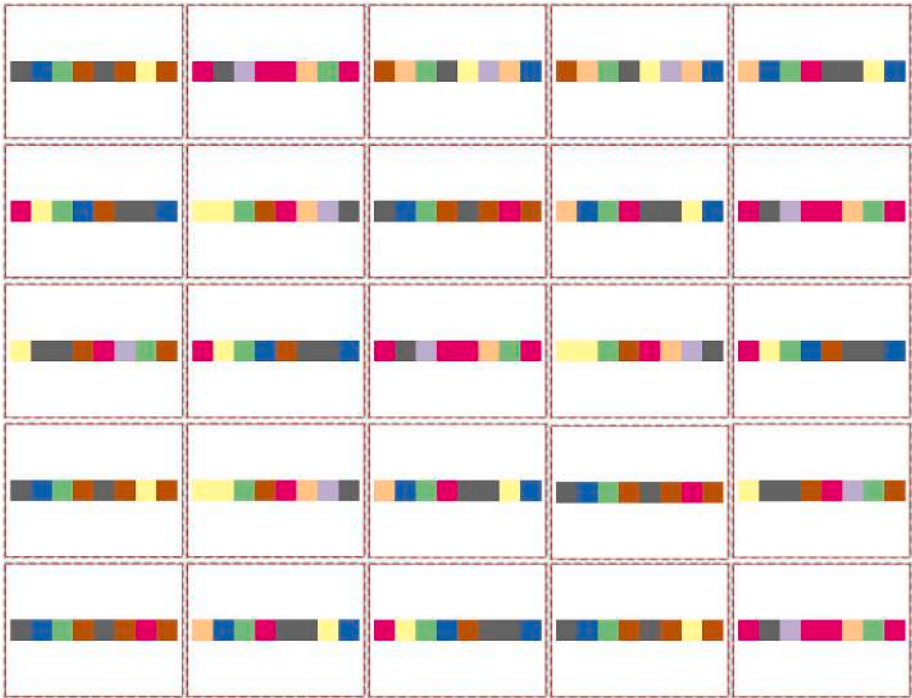


Fig. 27. Visualization of Output Features from the FC Block.

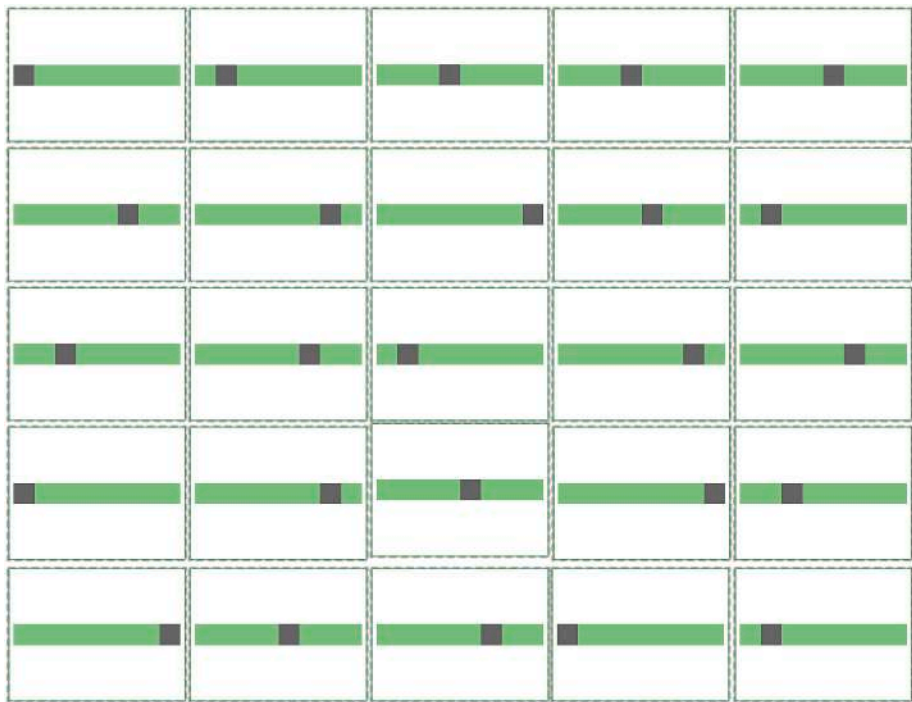


Fig. 28. Visualization of Output Features from the FC Block After Softmax.

Table 7

Comparison of the effectiveness of the proposed strategy under different proportions of sick broilers. The accuracy here refers to the percentage of the sick broilers' coughing frequency, as statistically recorded by the proposed strategy, compared to the actual coughing frequency manually labeled.

Area ID	Proportion of sick broilers in the sub-area	Actual coughing count (Manually labeled)	Coughing frequency recorded by the proposed strategy	Accuracy
1	3 %	14	14	100 %
2	6 %	27	27	100 %
3	10 %	43	41	95.34 %
4	13 %	68	62	91.17 %
5	17 %	71	64	90.14 %

application strategy, we took the frequency of coughing sounds from sub-mature broilers as an example. In sub-areas with five different proportions of diseased broilers, we applied the strategy to record the actual coughing frequency over a 5-minute period, which was manually annotated. The recorded frequencies were then compared with the coughing frequencies that were detected by the proposed strategy during the alert. The standard for coughing in this experiment was set at 10 occurrences. It is important to note that the proportion of diseased broilers in these five sub-areas was a natural result of the environment, not artificially set or intervened. Moreover, after the experiment, all identified diseased broilers were immediately isolated and treated effectively by veterinary husbandry experts. The results presented in Table 7 indicate that when the proportion of diseased broilers in the monitoring area is low, with only a few diseased broilers, the proposed strategy can accurately capture nearly all coughing events. Even as the proportion of diseased broilers increases and coughing events become more frequent, the accuracy of the strategy remains high (over 90 % in all cases), which strongly demonstrates the feasibility and robustness of the proposed strategy as an abnormal event monitoring system.

However, it should be acknowledged that, as shown in the experimental results in Table 7, the accuracy of the strategy decreases as the proportion of sick chickens in the breeding sub-area increases. This is

due to the overlapping sounds from multiple sick chickens, which to some extent drown out the features of different types of sound signals, negatively impacting the monitoring results. In ongoing research, we have proposed an evaluation metric to measure the extent to which these features are submerged, which we call the "submersion ratio." Additionally, using convolutional neural networks, we have introduced a novel feature extraction method and a broiler monitoring framework that adaptively adjusts the weight constraints when representing sound signals based on different submersion ratios, in order to improve the accuracy of broiler health monitoring under varying submersion conditions. We will share more detailed experimental procedures and reliable conclusions in future publications.

In fact, broilers infected with bronchitis also exhibit other visual clinical symptoms, such as abnormally moist eyes and abnormal feces color (Zhu et al., 2013). However, although visual information can provide unique pathological features, models based on visual data typically require higher computational overhead during both training and deployment compared to those based on audio signals. Therefore, how to effectively utilize the complementary nature of these two modalities in terms of pathological features, while reasonably controlling computational costs, to achieve relatively accurate monitoring performance with a multimodal approach for broiler chicken health monitoring is a topic worthy of further exploration in future research.

4. Conclusions

This paper proposes a novel Broiler Chicken Sound Signal Recognition Network (BSRNet), designed to enhance the accuracy of broiler chicken health monitoring or sound signal recognition without the need for cumbersome preprocessing of the original sound signals (such as filtering, denoising, and feature selection).

The core components of BSRNet include Separated Convolution (SepConv Convolution), Adaptive Pre-Activation Convolution (APA-Conv Convolution), and Dual Path Feature Fusion (DFF) structure. SepConv is aimed at enabling the network to autonomously identify key core information, dynamically enhancing attention to core features while diminishing focus on redundant, irrelevant, or noisy features.

APACnv can perform adaptive nonlinear mapping based on the characteristics of various local sound patterns captured by convolution kernels, thereby improving the network's ability to express complex nonlinear relationships. The DFF structure utilizes these novel convolutions and transforms feature channel dimensions (expansion and compression) to achieve multilevel feature fusion and abstract expression. Ablation studies have proven the effectiveness of these core components.

Comparative experimental results demonstrate that BSRNet outperforms all compared methods in recognizing broiler chicken sound signals, validating the effectiveness and advancement of the proposed method. The authors also conducted a series of interpretability analysis experiments using T-SNE, Grad-CAM++, and feature visualization, proving that BSRNet can dynamically adjust its focus on useful sound features and effectively distinguish key information across different signal categories.

In conclusion, the methods proposed in this paper achieve accurate monitoring of broiler chicken health, enhancing management efficiency during breeding and providing valuable technical references for sustainable green agriculture as well as the allocation of resources like feed and medication by breeders. Moreover, the methods can be extended to other similar sound signal recognition tasks, such as animal health monitoring or environmental noise classification, offering relevant insights for related fields.

Future research will focus on addressing the issue of feature submersion caused by the sounds of multiple sick broiler chickens. Additionally, multimodal broiler health monitoring technology will be developed by incorporating the clinical visual symptoms of sick chickens. Furthermore, the research will be expanded to include more comprehensive data, covering additional growth stages of the chickens. Consideration will also be given to the integration of diversified acoustic features extracted using different signal processing algorithms, in a reasonable manner, to further enhance the model's generalization ability. At the same time, the scope of the research will be broadened to

evaluate and validate the effectiveness and applicability of the proposed method in other sound signal recognition domains.

CRediT authorship contribution statement

Xiangjiang Zhao: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Zhihao Zhang:** Software, Resources, Project administration. **Chao Wang:** Visualization, Validation, Data curation. **Guotao Wang:** Writing – review & editing, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Shouli Song:** Visualization, Validation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1

Ablation experimental results on the RJ1 dataset.

Methods	Acc (%)	MP (%)	MR (%)	MF1 (%)
Our	66.93	88.63	84.37	79.95
Our-S	53.26	54.16	39.95	38.13
Our-S-P	58.66	67.45	50.46	66.78
Our-A	43.47	64.65	39.01	32.48

Table A2

Ablation experimental results on the RJ2 dataset.

Methods	Acc (%)	MP (%)	MR (%)	MF1 (%)
Our	98.72	100	100	100
Our-S	94.48	92.57	96.42	94.45
Our-S-P	96.22	94.56	96.99	96.34
Our-A	83.59	77.77	79.79	76.38

Table A3

Ablation experimental results on the RJ3 dataset.

Methods	Acc (%)	MP (%)	MR (%)	MF1 (%)
Our	99.02	100	100	100
Our-S	95.77	90.12	95.15	93.23
Our-S-P	96.55	93.46	96.45	95.33
Our-A	78.51	73.09	76.67	64.02

Table A4

Ablation experimental results on the RJ4 dataset.

Methods	Acc (%)	MP (%)	MR (%)	MF1 (%)
Our	61.12	75.29	79.79	77.87
Our-S	51.12	54.12	40.16	50.13
Our-S-P	55.32	56.76	60.88	62.45
Our-A	43.05	47.91	40.91	24.57

Table A5

Experimental results of performance comparison with eight methods on the RJ1 dataset.

Methods	Acc (%)	MP (%)	MR (%)	MF1 (%)
Our	66.93	88.63	84.37	79.95
LiConformer	63.46	84.84	82.29	78.46
MWANet	45.90	62.13	59.33	54.37
CapsuleNet	63.28	84.67	82.22	74.61
GTEFNet	62.48	65.40	54.58	53.34
LSTM	57.84	68.91	58.33	57.98
Transformer	38.82	44.37	34.08	30.14
RF	30.01	31.83	32.36	24.43
SVM	34.95	57.23	39.44	35.29

Table A6

Experimental results of performance comparison with eight methods on the RJ2 dataset.

Methods	Acc (%)	MP (%)	MR (%)	MF1 (%)
Our	98.72	100	100	100
LiConformer	90.39	85.41	88.33	81.53
MWANet	69.67	80.25	82.87	79.54
CapsuleNet	93.09	94.79	91.66	90.82
GTEFNet	88.56	64.84	88.02	90.77
LSTM	88.23	93.21	90.62	89.96
Transformer	76.37	67.60	72.39	61.54
RF	73.18	76.75	73.11	73.96
SVM	74.86	81.91	75.55	73.85

Table A7

Experimental results of performance comparison with eight methods on the RJ3 dataset.

Methods	Acc (%)	MP (%)	MR (%)	MF1 (%)
Our	99.02	100	100	100
LiConformer	89.97	95.37	94.44	94.72
MWANet	72.40	80.67	85.03	79.86
CapsuleNet	93.02	97.51	95.83	96.11
GTEFNet	89.17	95.83	97.52	96.11
LSTM	89.78	94.28	93.45	93.01
Transformer	77.20	71.73	57.05	56.38
RF	76.93	77.27	74.25	75.11
SVM	76.58	82.74	77.62	77.21

Table A8
Experimental results of performance comparison with eight methods on the RJ4 dataset.

Methods	Acc (%)	MP (%)	MR (%)	MF1 (%)
Our	61.12	75.29	79.79	77.87
LiConformer	46.07	52.67	33.09	27.57
MWANet	39.14	72.37	54.78	48.36
CapsuleNet	58.72	67.55	61.01	58.15
GTEFNet	56.82	61.25	47.47	49.37
LSTM	48.34	41.49	43.97	40.89
Transformer	37.25	43.59	26.19	20.76
RF	44.51	43.00	45.28	40.02
SVM	52.55	67.14	43.08	44.76

Data availability

Data will be made available on request.

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